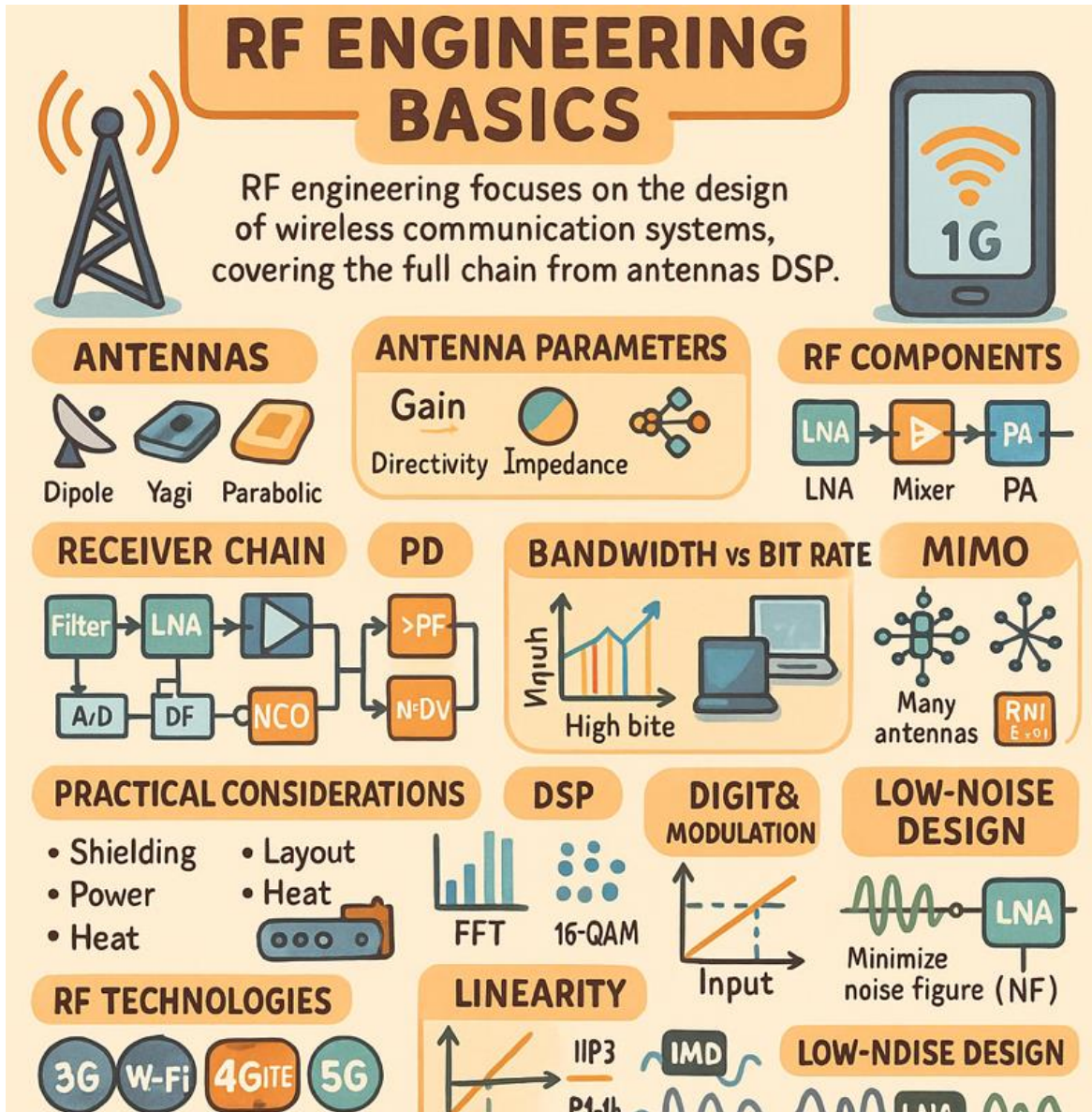




RF Circuit Design Basics

From Basics to 5G

Agenda, what we will learn

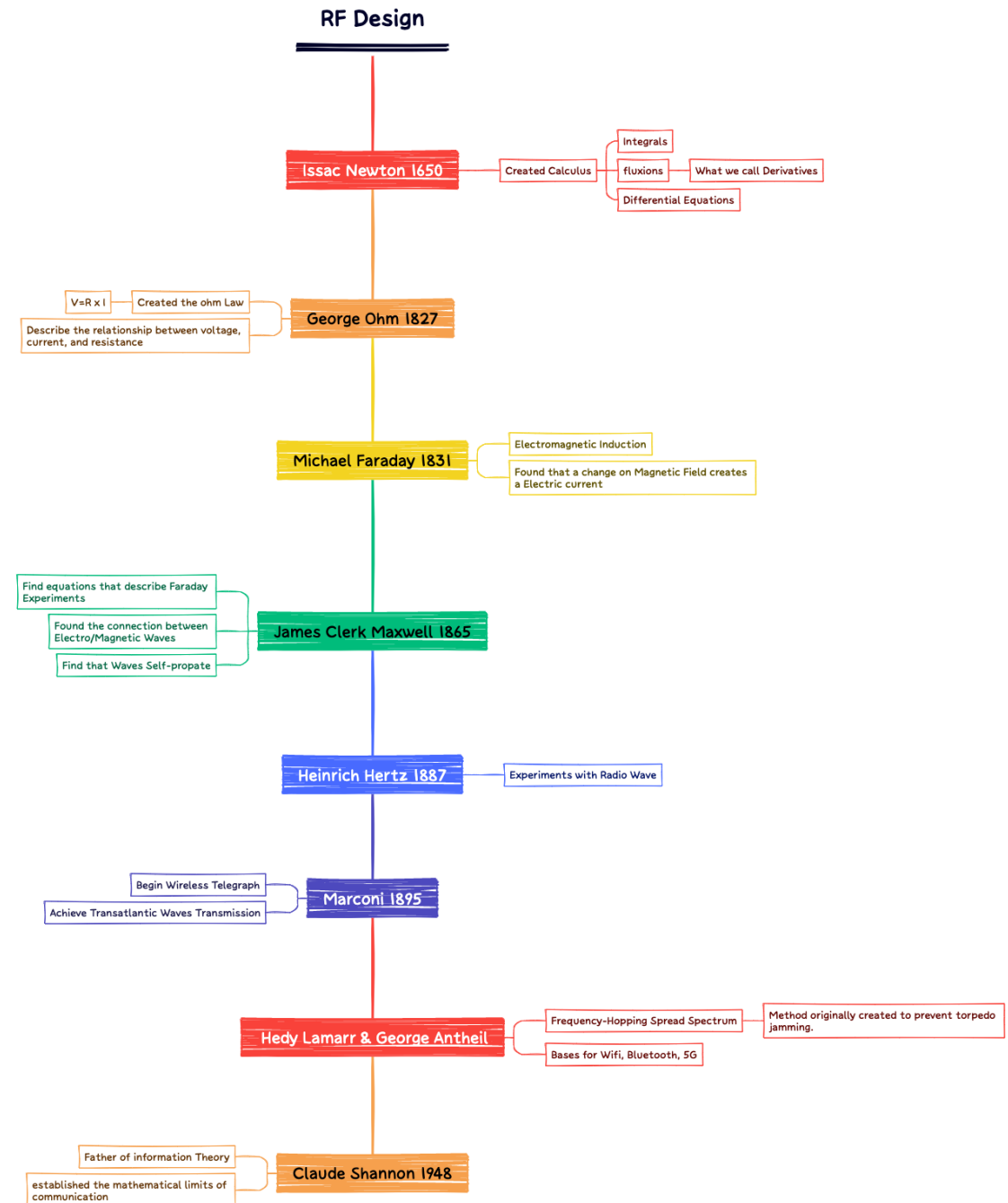
-  **Foundations of Electromagnetism**
 - Maxwell's Equations (in simple terms)
 - Faraday, Lorentz, Hertz, Marconi, Lamarr
 - What EM waves are & how antennas radiate
 - Why EM waves detach from circuits
-  **Signals, Frequency & Bandwidth**
 - Carrier frequency vs bandwidth
 - Why bandwidth = information speed
 - Channels inside frequency bands
 - Shannon Capacity (simple version)
 - Modulation basics (AM, FM, QAM, OFDM)
-  **Real RF Components**
 - Oscillators & Local Oscillators (LO)
 - Mixers: up conversion & down conversion
 - Filters (LPF, BPF, SAW, BAW)
 - Amplifiers: LNA, PA
 - Noise, SNR & Noise Figure (easy)
 - Antenna basics ($\lambda/2$, impedance, matching)

-  **Receiver & Transmitter Architectures**
 - Superheterodyne Receiver (Downconversion)
 - Superhet Transmitter (Upconversion)
 - Intermediate Frequency (IF) explained
 - Zero-IF / Direct Conversion (modern radios)
 - I/Q signals & complex baseband
 - Sampling rate vs bandwidth (Nyquist)
-  **Spectrum & Wireless Standards**
 - FM, AM, Bluetooth, Wi-Fi, LTE, 5G bands
 - Low-band, mid-band, mmWave comparison
 - Channel widths & regulatory constraints
-  **Software-Defined Radio (SDR)**
 - What SDR is and why it matters
 - ADC/DAC and digital down conversion
 - Baseband DSP (filters, FFT, demodulation)
 - Example SDR pipeline: antenna $\rightarrow$ bits



Origins of RF

These are the people responsible to start the field of Radio Frequency, we will investigate their work and how to apply on our projects.



Isaac Newton (The Inventor of Calculus)

What Newton Created

Newton (together with Leibniz) created **calculus**, the mathematical framework that describes:

- change over time (**derivatives**)
- accumulation of quantities (**integrals**)
- [Differential Equations](#)
- motion, forces, and fields in continuous space

This became the foundation of all later physics — including electromagnetism.

Derivatives — measuring change

Maxwell's equations rely on **how fields change in time**:

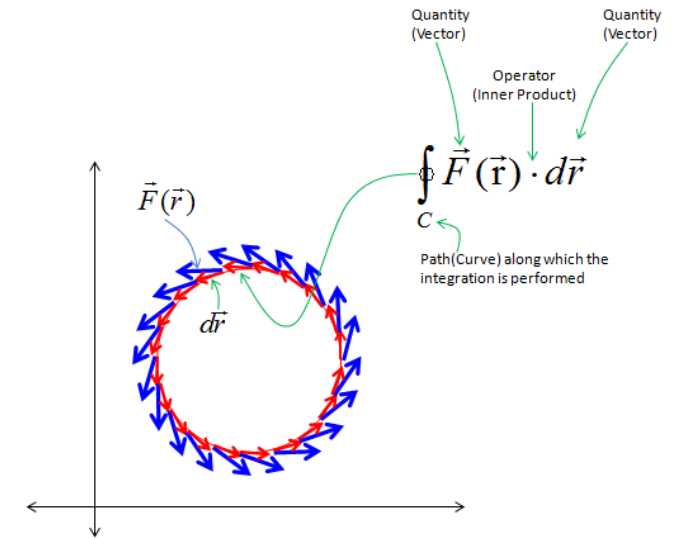
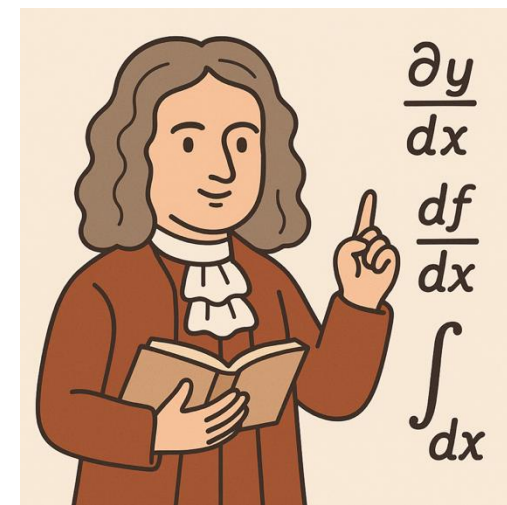
$$\frac{\partial E}{\partial t}, \frac{\partial B}{\partial t}$$

Integrals — measuring accumulation

Maxwell uses integrals to describe how fields accumulate across:

- **surfaces** (flux)
- **closed loops** (circulation), [closed loop integral](#), which is different than a integral from point a to b, on this case we go from a to b, then back to a

$$\oint \mathbf{E} \cdot d\mathbf{l}$$



Grad, Div, Curl

Divergence: Measure of Source/Sink

- Positive divergence → field lines radiate outward (a source)
- Negative divergence → field lines converge inward (a sink)

Curl: Measure of rotation around a point.

- Why This Matters:

Maxwell's equations are written using divergence and curl. If you don't know these operators, the equations look like cryptic symbols.

With divergence and curl:

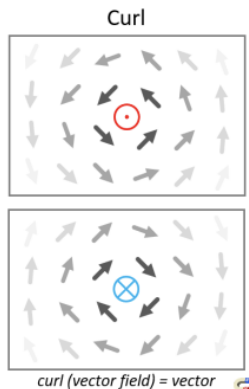
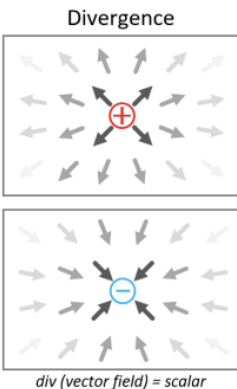
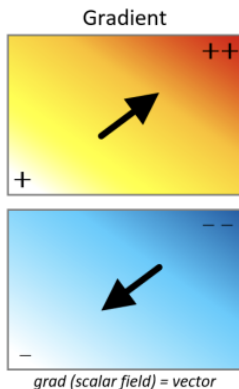
- Gauss's Law becomes "charges create field spread"
- Gauss's Law for Magnetism becomes "magnetic field has no sources"
- Faraday's Law becomes "changing B makes swirling E"
- Maxwell–Ampère Law becomes "currents + changing E make B"

	Cartesian (x, y, z) $\mathbf{i}, \mathbf{j},$ and $\mathbf{k}$ are unit vectors in the directions of increasing $x, y,$ and z . $M, N,$ and P are the scalar components of $\mathbf{F}(x, y, z)$ in these directions.
Gradient	$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$
Divergence	$\nabla \cdot \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}$
Curl	$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix}$
Laplacian	$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$

Vector Triple Products

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u} = (\mathbf{w} \times \mathbf{u}) \cdot \mathbf{v}$$

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \cdot \mathbf{w})\mathbf{v} - (\mathbf{u} \cdot \mathbf{v})\mathbf{w}$$



George Ohm (1825-1827)

How the first circuit law was discovered

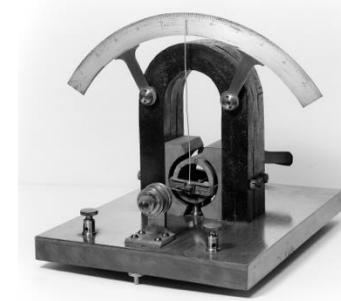
Before Ohm (early 1800s)

- Electricity had sparks and batteries, but no way to measure current
- No known relationship between voltage (“pressure”) and current (“flow”)
- No quantitative theory of circuits

Ohm’s Experiments

- Using one of the first accurate **galvanometers**, he systematically varied:
- Wire **length**, **diameter**, and **material**
- Battery **voltage**

Ohm’s Discovery From the data he organized the results in a table.

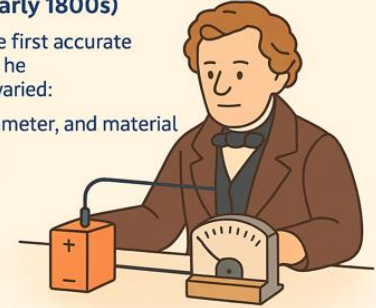


Ohm’s Breakthrough (1825–1827)

How the first circuit law was discovered

Before Ohm (early 1800s)

- Using one of the first accurate galvanometers, he systematically varied:
- Wire length, diameter, and material
- Battery voltage



Ohm’s Experiments

- Using one of the first accurate galvanometers, he systematically varied:
 - ↑ Wire length, diameter, and material
 - ↑ Battery voltage
- ↑ Voltage → ↑ Current
↑ Thickness → ↑ Current
Different metals → different conductivities

Ohm’s Discovery

Experimentally derived:

$$\frac{I}{V} = \text{rewritten as: } I = \frac{V}{R}$$

Ohm was the first to quantify how circuits behave – purely by experiment.

helandessi.com

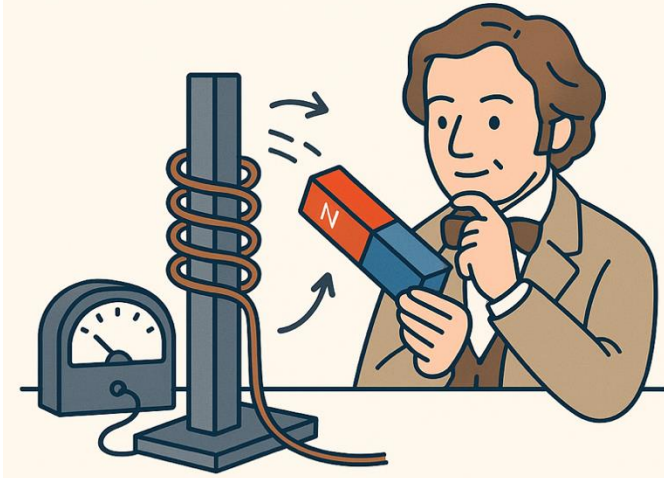


Voltage V (volts)	Current I (amperes)	
Experiment	I/V	
1	0,12	0,12
2	0,24	0,36
3	0,36	0,48
4	2,0	0,48

Michael Faraday

Faraday had not Much math skills but he empirically discovered that a magnetic field is created When we pass a current into a wire, and also When we move a magnetic into into a coil it also introduce some current.

Faraday discovering induction



Moving magnet induces
current in coil

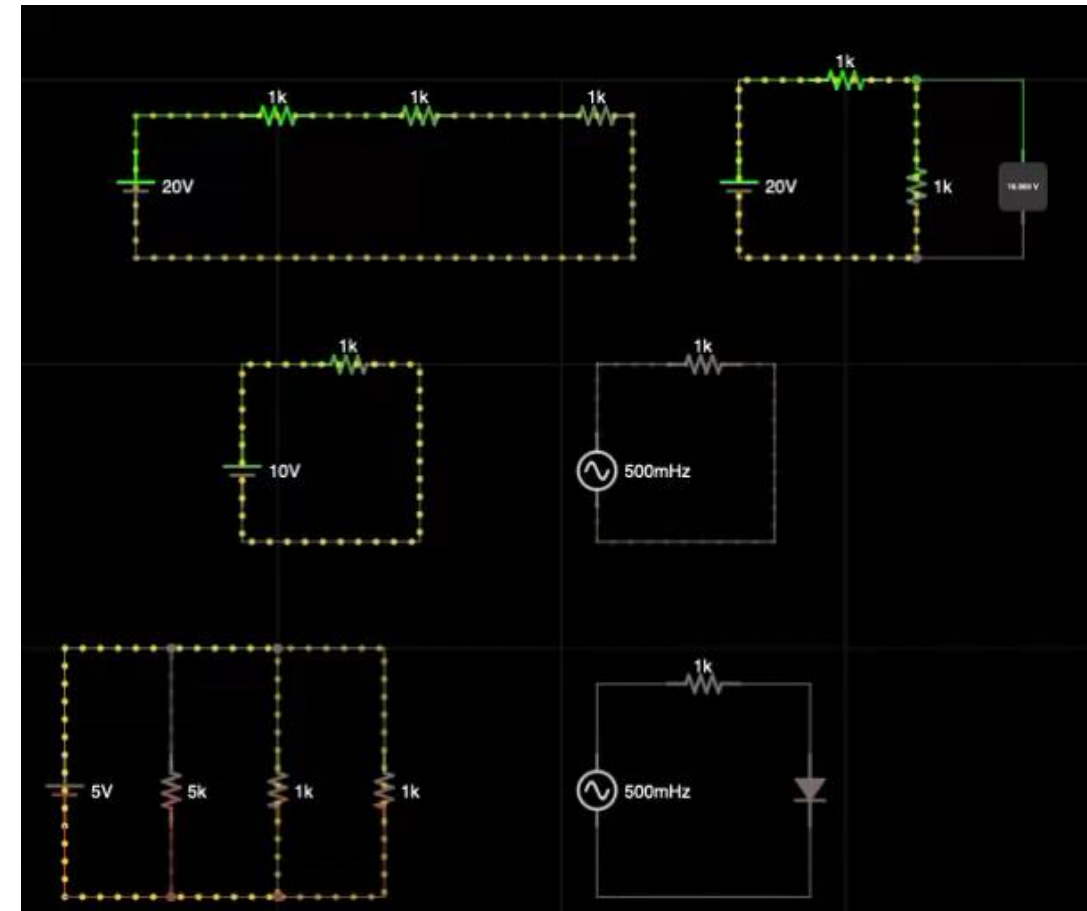
Before Maxwell

Many people imagine that electrons in a circuit move at the speed of light.

But that isn't true. Electrons drift extremely slowly — only a few tenths of a millimeter per second, which is less than one kilometer per hour.

What actually travels near the speed of light is the electromagnetic field produced by the circuit. When voltage is applied, the electric field changes almost instantly along the wire, and this field pushes the electrons.

So the signal — the information and energy — moves at light speed, while the electrons themselves crawl forward very slowly.



Maxwell Equations

He basically “interpolated” the results he saw in Faraday’s experiments and also

formalized them using the mathematical tools developed since Newton and Leibniz.

By combining experimental laws (Coulomb, Ampère, Faraday) with the language of

calculus, Maxwell unified electricity and magnetism into a single theory.

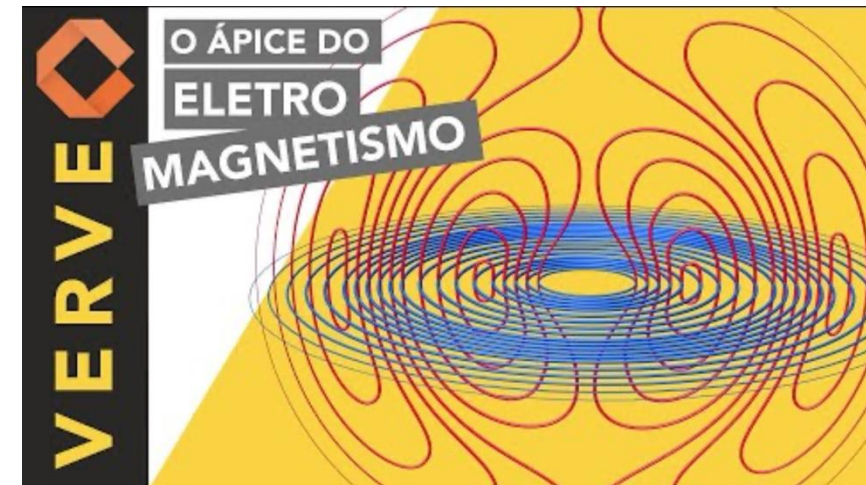
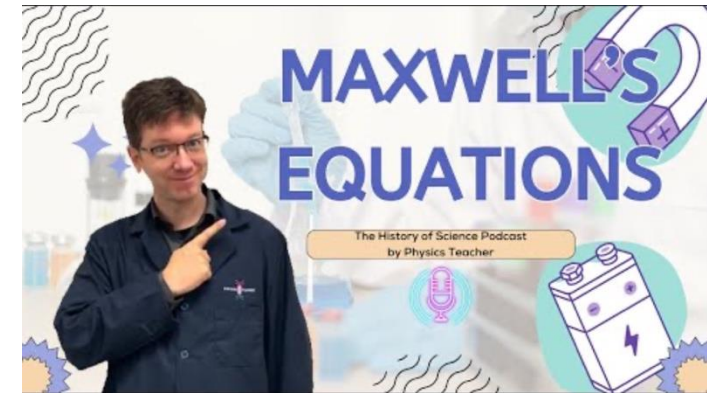
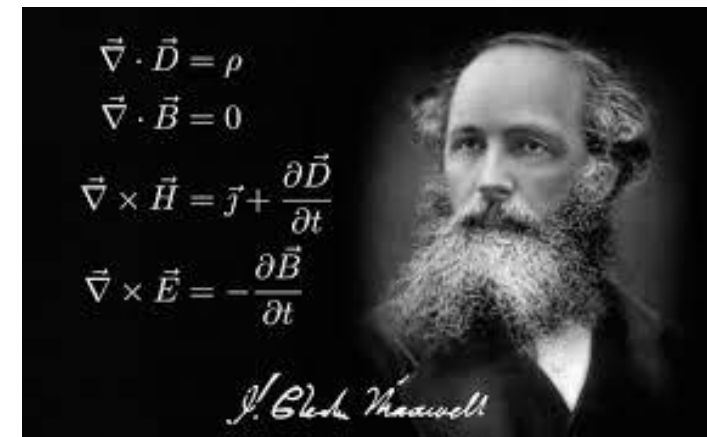
In doing so, he discovered something no one had seen directly before:

that changing electric and magnetic fields can sustain each other and

propagate through space as waves — electromagnetic waves.

This prediction is what later allowed Hertz to demonstrate radio waves

experimentally, and ultimately what made modern wireless communication possible.



Meaning of Each Term of the Equations

Maxwell equations are actually first order Partial differential equations with space and time dimensions

Gauss's Law (Electric Fields)

- Electric charges create electric fields.
- Positive charges “push” field lines outward; negative charges “pull” them inward.
- This explains static electricity and how voltage builds up.

Gauss's Law for Magnetism

- There are **no magnetic charges** (no monopoles).
- Magnetic field lines always form loops.
- Magnets always have a north AND south pole.

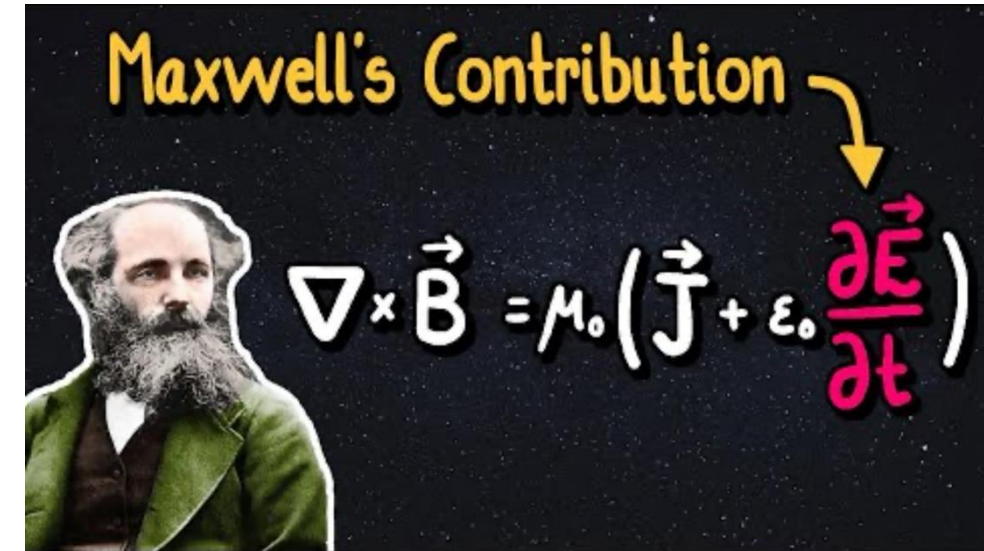
Faraday's Law of Induction

- A **changing magnetic field** creates an electric field.
- This is how generators, transformers, and inductors work.
- It explains why moving a magnet through a coil makes current flow.

Ampère–Maxwell Law

- Two things create magnetic fields:
- **electric current** (flowing charges)
- **changing electric field** (Maxwell's crucial addition)
- This is the law that **allows radio waves to exist**.

E and B can sustain each other and travel through space as an electromagnetic wave.



Name	Differential form	Integral form
Gauss's law	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$	$\oiint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{Q(V)}{\epsilon_0}$
Gauss's law for magnetism	$\nabla \cdot \mathbf{B} = 0$	$\oiint_{\partial V} \mathbf{B} \cdot d\mathbf{A} = 0$
Maxwell–Faraday equation (Faraday's law of induction)	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial \Phi_{B,S}}{\partial t}$
Ampère's circuital law (with Maxwell's correction)	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$	$\oint_{\partial S} \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_S + \mu_0 \epsilon_0 \frac{\partial \Phi_{E,S}}{\partial t}$

Lorentz Force and Electromagnetic fields



While the charges move into a direction a magnetic field is created perpendicular to the motion of the charges (follow the right-hand rule).

This interaction between moving charges and magnetic fields is captured by the Lorentz Force:

It tells us:

- The electric field E pushes charges in the direction of the field.
- The magnetic field B bends the path of moving charges at **right angles** to their velocity.
- The amount of bending depends on the speed v and the field strength B .

This is fundamental to RF because:

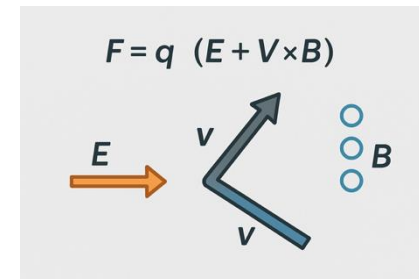
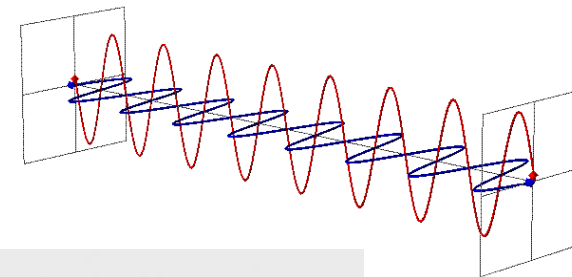
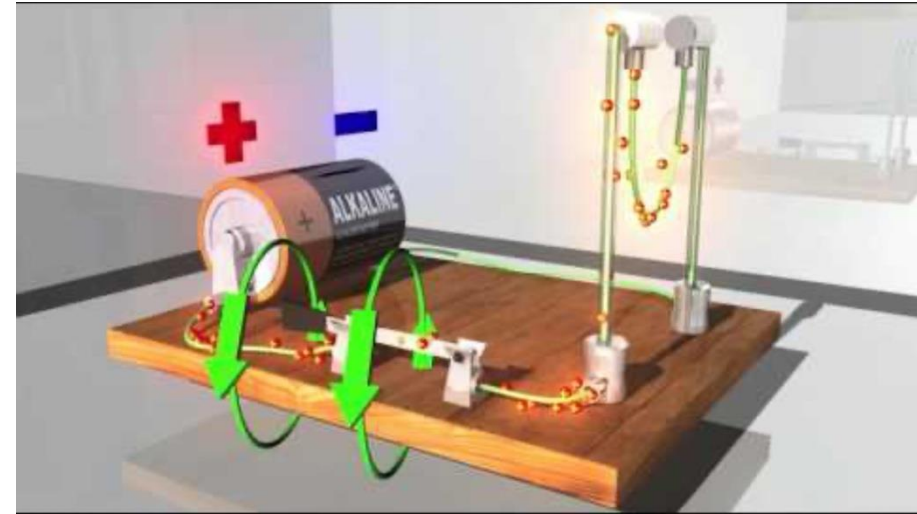
Oscillating charges $\rightarrow$ create oscillating magnetic fields.

Oscillating magnetic fields $\rightarrow$ create oscillating electric fields.

The E and B fields continually regenerate each other $\rightarrow$ forming an electromagnetic wave.

Together, Maxwell + Lorentz give the complete picture:

- Fields create motion of charges, and motion of charges creates fields.
- This feedback loop is what makes antennas radiate and what allows RF to propagate through space.



Hertz Experiment

What Hertz Set Out to Do

James Clerk Maxwell predicted electromagnetic waves in 1865 — but no one had ever *observed* them.

Hertz wanted to answer one question:

Do electromagnetic waves really exist?

The experiment:

Transmitter

- A high-voltage induction coil
- Two metal spheres with a small **spark gap**
- When the gap sparks → electrons oscillate → EM waves emitted

Receiver

- A loop of wire with a tiny gap
- If EM waves reach it → they induce a voltage → a visible microscopic spark jumps the gap

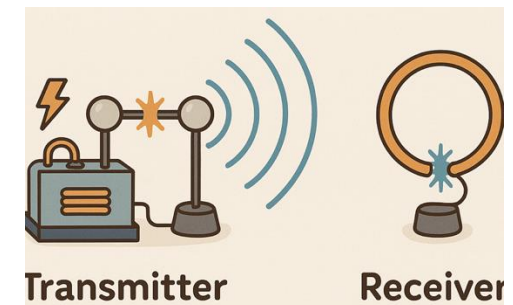
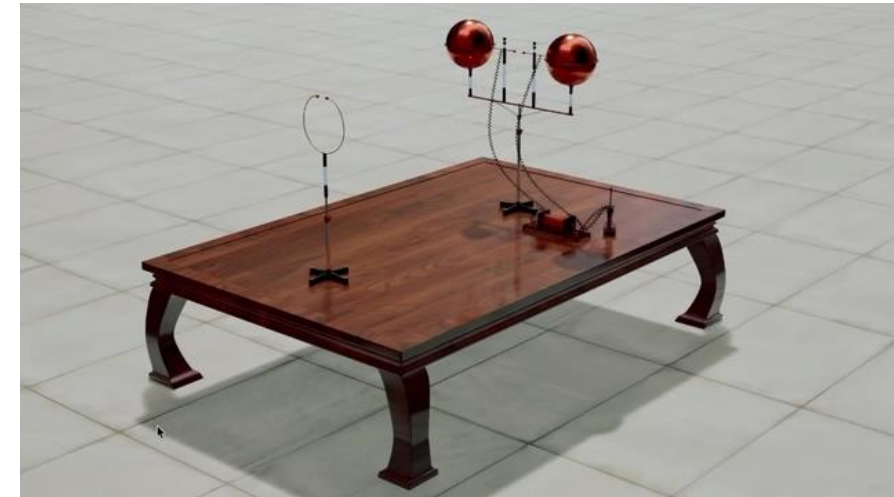
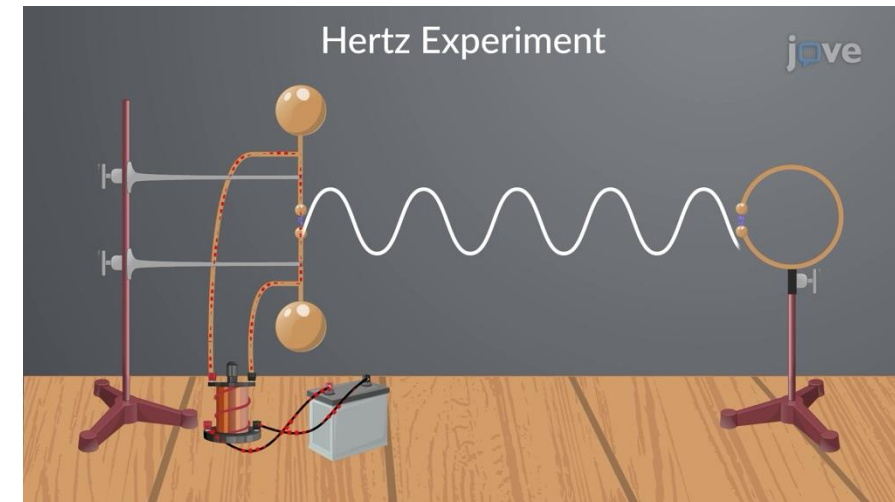
What Hertz Observed

- When the transmitter sparked, the receiver produced a spark *even across the room*
- Moving the receiver changed the spark intensity
- The spark disappeared when he blocked the path with metal sheets
- Waves could reflect, refract, interfere, and form standing waves

He had discovered **radio waves**.

What Hertz Proved:

- Maxwell was right, electromagnetic waves exist and move at speed of light
- Oscillating Charges radiate waves



Hedy Lamarr (1914–2000)

The Actress Who Invented Spread-Spectrum and Paved the Way for Wi-Fi, Bluetooth, and 5G

What She Did

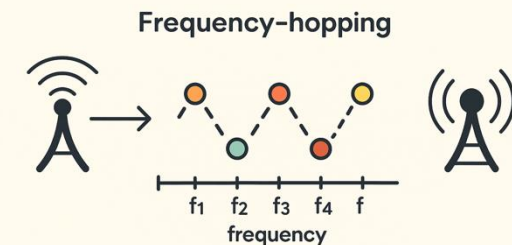
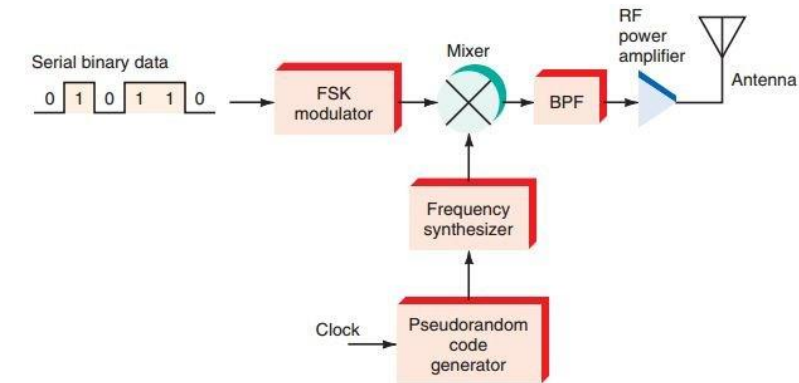
- Co-invented **Frequency-Hopping Spread Spectrum (FHSS)** in 1941
- Partnered with composer George Antheil
- Created a system where the transmitter and receiver **switch frequencies together** following a secret pattern
- Designed to prevent radio-controlled torpedoes from being jammed or intercepted during WWII

What Her Invention Solved

- Enemy forces could easily jam or block single-frequency radio signals
- Lamarr's hopping system spread the signal across many channels, making it:
 - hard to jam
 - hard to detect
 - hard to intercept
- This made wireless communication **secure and resilient**

The idea is that the transmitter change it's frequency over time, following a pattern, the receiver also keep selecting the frequency following the same pattern.

This way she found a way to avoid the enemies to Jam the communications.



Claude Shannon

Claude Shannon (1916–2001) was an American mathematician and engineer who created the entire field of **digital communication**. He is to information theory what **Maxwell** is to electromagnetism.

His Big Idea (1948)

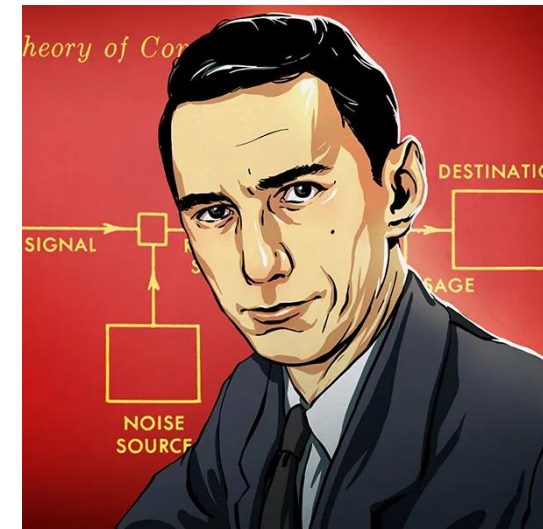
- In his landmark paper “*A Mathematical Theory of Communication*”, Shannon:
- Invented the concept of the **bit**
- Introduced **entropy** as a measure of information
- Defined what a **channel** is
- Defined what **noise** is
- Discovered the **maximum data rate** any channel can carry

Why Shannon Matters in RF

Shannon's theory explains:

- Why **more bandwidth** → more speed
- Why **cleaner signals (higher SNR)** → more bits per symbol
- Why modern systems use **QAM, OFDM, error-correction codes**
- Why no radio (Wi-Fi, LTE, 5G, Bluetooth) can exceed his limit

Every wireless standard in existence is built on Shannon's ideas.



The Engineer Who Made Wireless Communication Real

What Marconi Did

- Built the **first practical radio transmitter** (spark-gap + tuned circuits)
- Invented the **first sensitive radio receiver** (coherer detector)
- Discovered that **tall antennas + good grounding** dramatically increase range
- Demonstrated the **first long-distance wireless links**
- 1895: kilometers
- 1899: across the English Channel
- **1901: first trans-Atlantic signal ("S")**

What He Discovered

- Radio waves can travel far beyond line-of-sight
- Resonance (tuned circuits) makes transmission more efficient
- Weak RF signals can be detected reliably with proper circuitry
- Wireless telegraphy is possible over oceans and continents

His Contribution to RF

- Turned Maxwell's theory + Hertz's sparks into a **working communication system**
- Laid the foundation for antennas, receivers, transmitters, and propagation studies
- Launched the first wireless company and the entire field of radio communication

THE ENGINEER WHO MADE WIRELESS COMMUNICATION REAL

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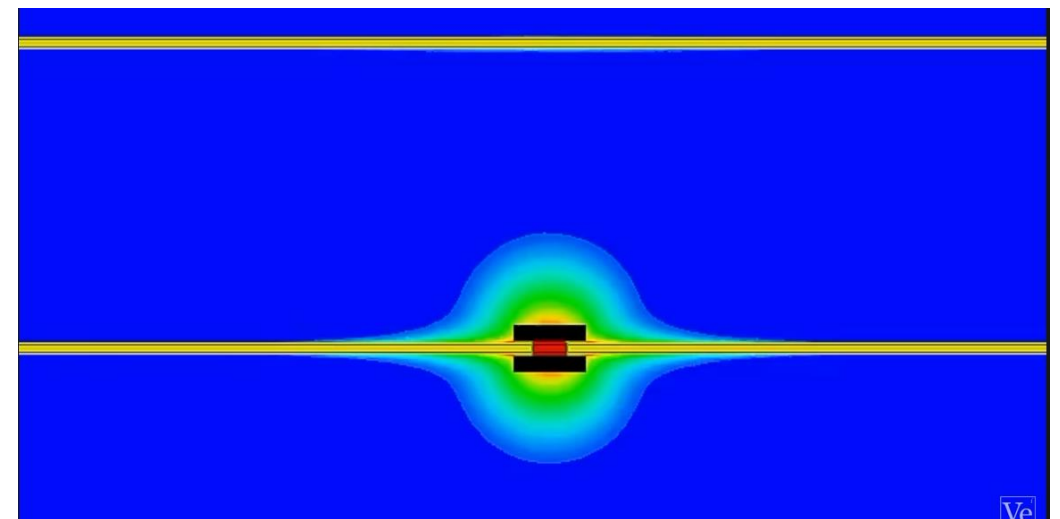
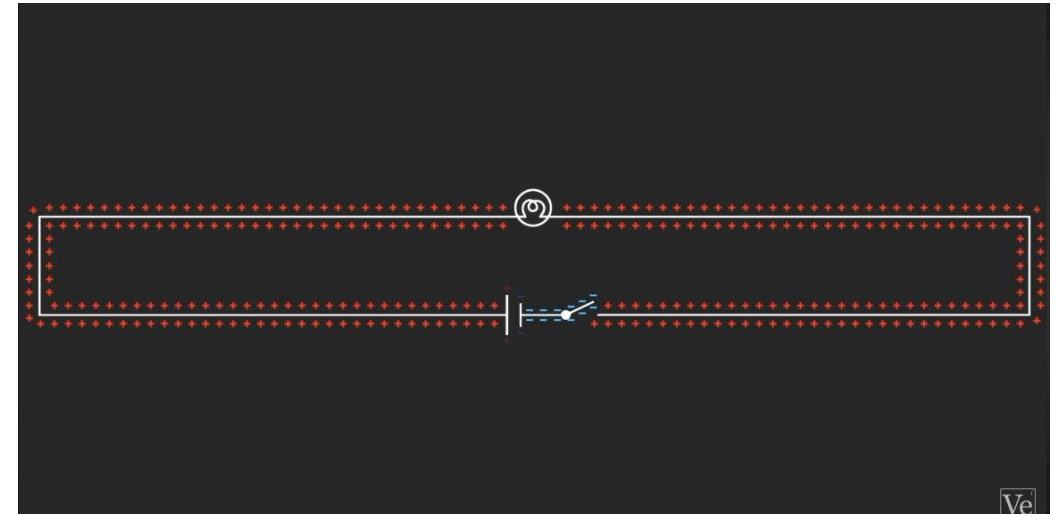


So how the circuits really works?

So, we know that is not exactly the electrons that runs at speed of light, what happens is an electromagnetic field that travels at the speed of light from the point Where the circuit is closed (i.e.: switch) to the passive element (i.e.: a Lamp) and this electromagnetic field that travels at the speed of light.

Actually, even if we have a disconnected wire from the circuit, it will be influenced by the electromagnetic field and produce a current.

But the truth is that we don't design circuits thinking all the time on maxwell equations, if the Frequency is not high enough, we can still think as electric circuits.



How does an EM wave propagate away from an antenna?

When a circuit drives an antenna with a rapidly alternating current, the electrons inside the conductor move back and forth. This motion creates **rapidly changing electric and magnetic fields** around the wire.

At the **ends of the antenna**, the electrons cannot continue forward because there is no more conductor. They momentarily build up and reverse direction.

This abrupt stop-and-reverse creates strong, changing fields just outside the metal.

These changing fields are the key:

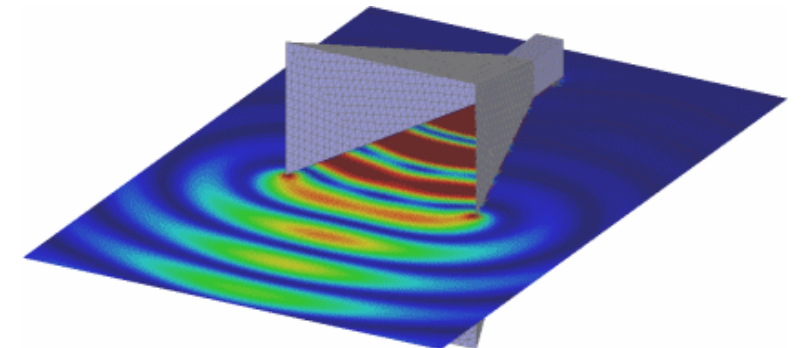
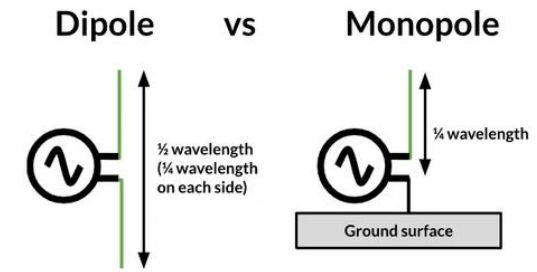
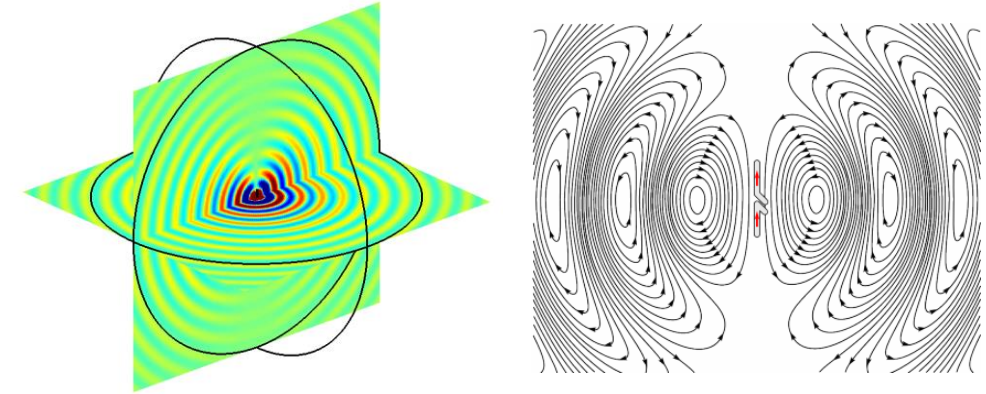
- A **changing electric field** creates a magnetic field
- A **changing magnetic field** creates an electric field

This feedback loop allows the fields to **sustain each other**.

Once this loop forms near the antenna tips, it **no longer depends on the wire**. The fields “detach” and continue traveling outward as a self-propagating electromagnetic wave.

In simple terms:

- The antenna shakes the electric and magnetic fields,
- and those fields shake each other as they move away.





How the Antenna will capture EM?

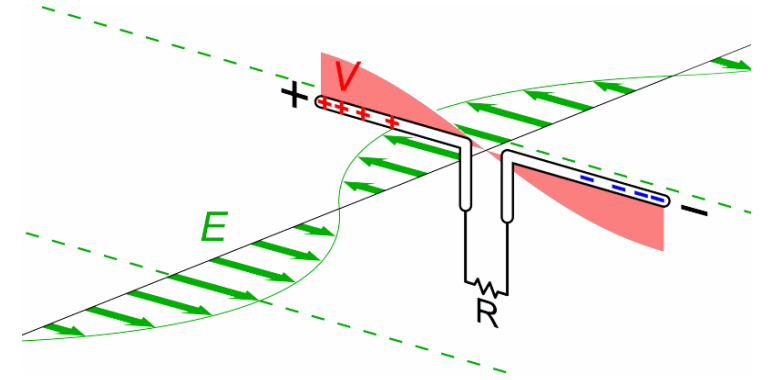
An incoming electromagnetic wave makes the electrons inside the antenna wire **move back and forth**.

The electric field of the wave pushes the charges, creating a tiny alternating voltage and current in the metal.

This AC signal travels into the circuit, where it can be amplified, filtered, and decoded.

In simple terms:

- the EM wave “shakes” the electrons in the antenna, and that motion becomes electricity.





Antenna Length & Wavelength

Antennas work by coupling to electromagnetic waves.

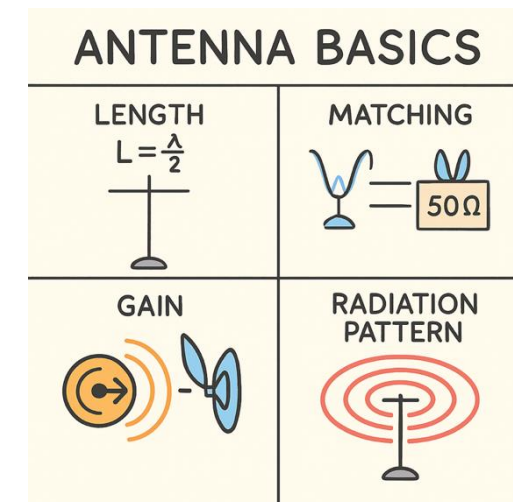
For a simple **half-wave dipole**: $L = \frac{\lambda}{2}$, and $\lambda = \frac{c}{f}$

Higher **frequency** → shorter **antenna**.

Also on our projects we need to match antenna impedances, to ensure maximum power transfer from the transmitter to the antenna.

We will use techniques like Smith Chart to match the impedance between our circuit and the antenna.

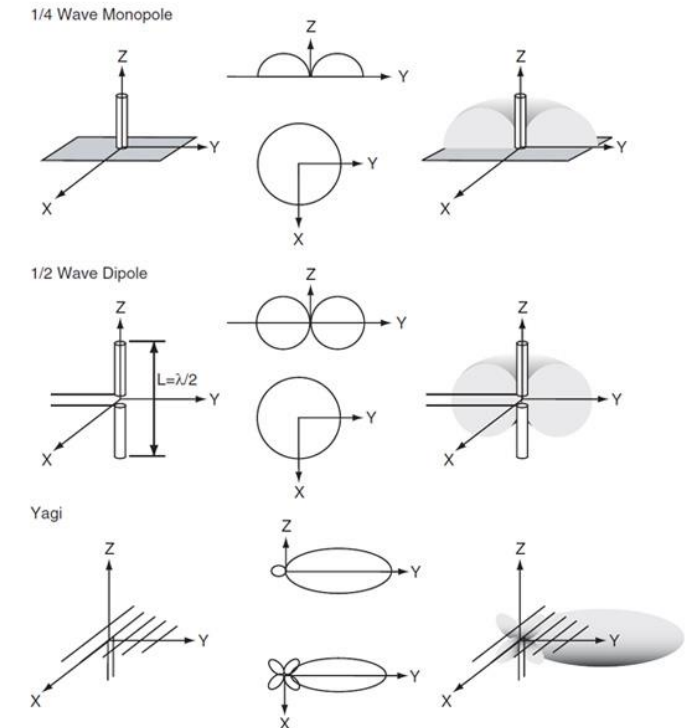
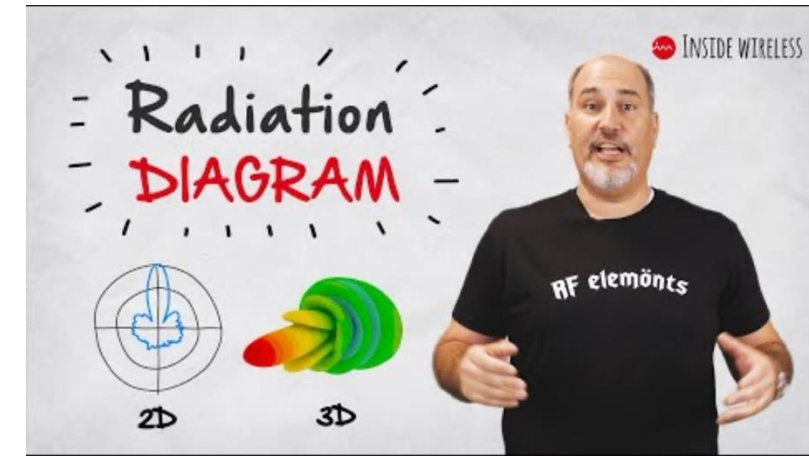
On the next slide we will learn about antenna types and their radiation pattern.



Frequency	Wavelength	λ/2 Dipole Length
100 MHz	3 m	1.5 m
2.4 GHz	12.5 cm	6.25 cm
28 GHz	1.07 cm	5.3 mm

Antenna Radiation Pattern

Antenna Type	Radiation Pattern Shape	Pattern Description	Typical Use
Half-wave Dipole	🍩 Donut-shaped (torus)	Strong broadside, nulls at the ends	Wi-Fi, broadcast, general RF
Monopole ($\lambda/4$)	🗺️ Half-donut over ground plane	Omni in horizontal plane, null at top	Phones, routers, IoT
<u>Patch</u> (Microstrip)	📶 Directional “pancake”	Moderate beamwidth, front-directed	Wi-Fi APs, 5G phones, GPS
Yagi-Uda	➡️ Narrow front-facing beam	Very directional, high gain	TV antennas, long-distance links
Parabolic Dish	🔭 Very narrow pencil beam	Highest gain, extremely focused	Satellite, radar, microwave links
Horn Antenna	🔊 Conical directional beam	Medium gain, controlled beam	Microwave, radar, 5G mmWave
Helical Antenna	🌀 Circular polarization beam	Axial mode → directional spiral	Satellite comms, GNSS
Loop Antenna	🔄 Figure-8 pattern	Weak at center, strong at sides	AM receivers, RFID
Array / Phased Array	🎯 Steerable beam	Beam electronically steered	5G Massive MIMO, radar
Log-Periodic Antenna	📏 Wideband directional	Long-range with wide frequency range	Spectrum analyzers, HF/VHF



Path Loss & Propagation Basics

Wireless signals get weaker as they travel through space. This reduction in signal strength is called **path loss**.

Free-Space Path Loss (FSPL)

- Even in perfect empty space, a signal spreads out as a sphere.
- Power density decreases with distance:

Higher frequencies lose power faster:

- 3 GHz loses more than 900 MHz at the same distance.
- 28 GHz (5G mmWave) loses dramatically more.

Real-World Propagation

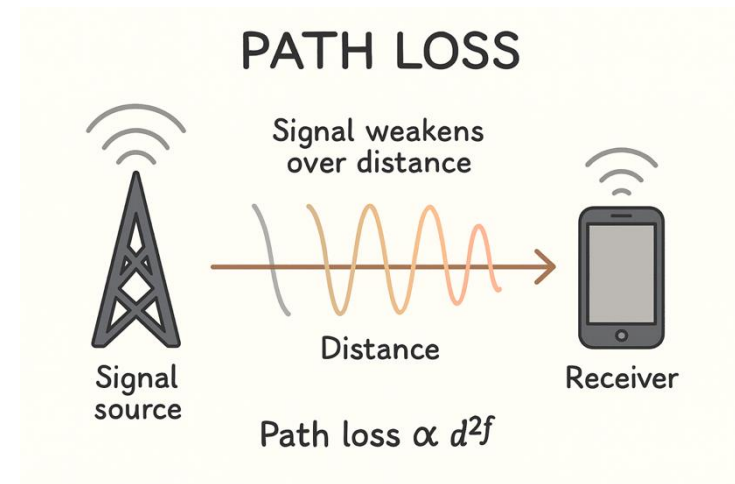
The environment adds even more attenuation:

Walls, floors, buildings → absorption + scattering

- Human body → strong attenuation (especially at 5G)
- Foliage → strong loss at mid/high frequency
- Multi-path → reflections, echoes, fading
- Diffraction → bending around corners (works at low freq)

Lower frequencies travel farther and penetrate better.

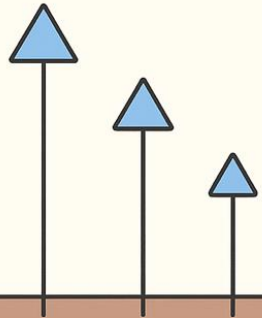
Higher frequencies need more antennas, beamforming, or repeaters.



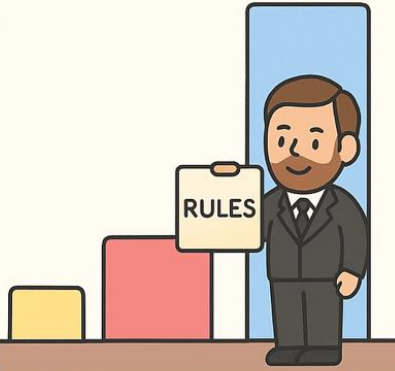
If low is better why use high frequencies?

CARRIER vs BANDWIDTH vs REGULATION

CARRIER FREQUENCY



BANDWIDTH



REGULATION



There Is Limited Low-Frequency Spectrum

Frequencies below 1 GHz are *extremely crowded*:

- FM radio
- TV broadcast
- Emergency radios
- Military
- IoT, GSM, ISM



Lower Frequency = Lower Bandwidth = Lower Data Rate

Available spectrum at low frequencies is narrow:

- At **700–900 MHz**, channels are typically **5–20 MHz**
- At **3.5 GHz**, channels are **50–100 MHz**
- **Reemember Shannon's Law $\text{DataRate} = B \cdot \log_2(1 + \text{SNR})$**

This doesn't mean a lower carrier frequency doesn't allow a high bandwidth, it's just the governments that doesn't allow a higher bandwidth.

- You **could** have a 1 GHz carrier with 500 MHz of bandwidth *if regulators allowed it*.
- But they don't — because that spectrum is full of legacy systems.

The *height* of the shelf (carrier frequency) does not tell you how *wide* the shelf is (bandwidth).

But newer, higher shelves are empty, so you can install bigger things on them.

Carrier Frequency & Data Rate

A **carrier frequency** is the high-frequency sinewave (RF tone) used to *carry* information over the air.
Examples:

- FM Radio → 100 MHz
- Wi-Fi → 2.4 GHz / 5 GHz
- 5G → 3.5 GHz / 28 GHz
- Bluetooth → 2.402–2.480 GHz



How Do We Put Information On It?

We **modulate** the carrier with data:

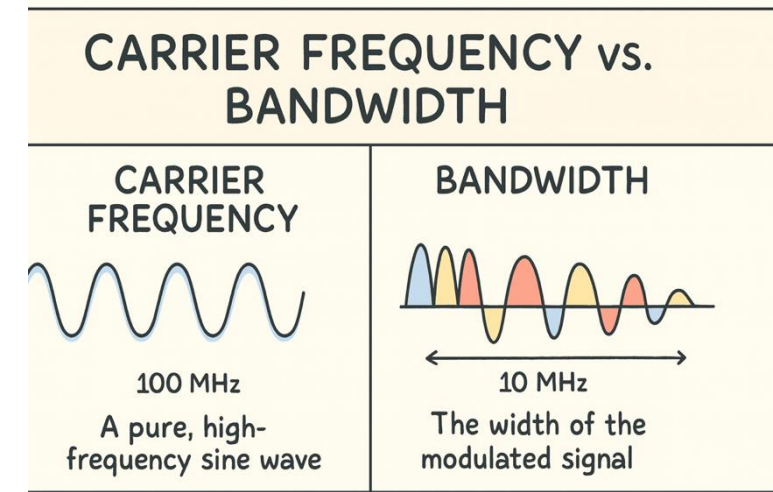
- amplitude changes (AM)
- frequency changes (FM)
- phase changes (PSK)
- amplitude + phase (QAM)
- multicarrier modulation (OFDM)



Carrier Frequency: What It Actually Controls

- How big the antenna must be
 - Higher frequency → smaller antenna
- How far the signal can travel
 - Lower frequency → longer range
 - Higher frequency → more attenuation
- How much bandwidth can be allocated
 - (High frequencies have more spectral space)
- Propagation characteristics
 - Lower freq → walls, obstacles penetration
 - Higher freq → short-range, directional

One important detail is that carrier frequency doesn't relate to data rate, the bandwidth does.



System	Carrier	Bandwidth	Speed
FM Radio	100 MHz	0.2 MHz	Slow (audio only)
Wi-Fi 2.4 GHz	2.4 GHz	20 MHz	~100 Mbps
Wi-Fi 5 GHz	5 GHz	80–160 MHz	500–1500 Mbps
5G mmWave	28 GHz	400–800 MHz	1–5+ Gbps

Modulation

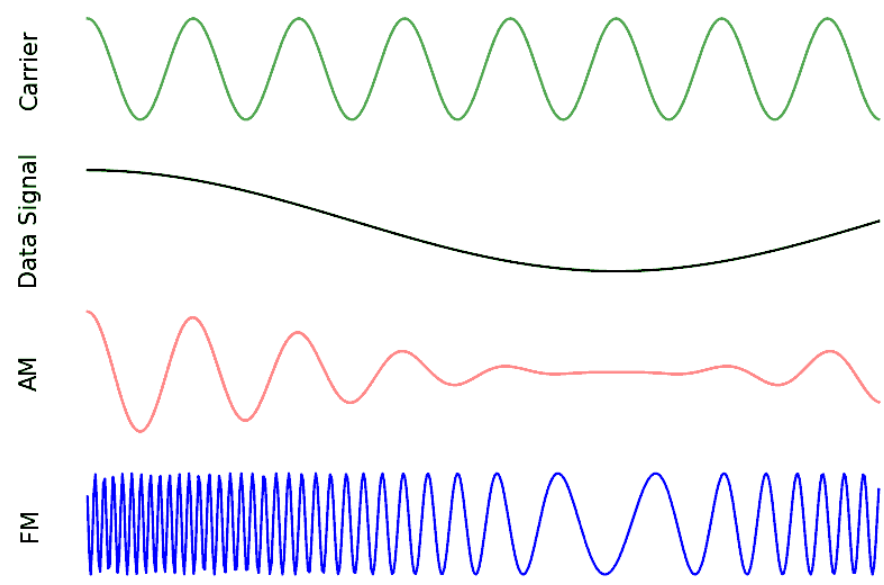
In order to “package” information onto a carrier, we need to modulate it.

Modulation means **changing some property** of a high-frequency carrier wave to represent data, sound, or symbols.

A carrier wave has three things we can change:

- **Amplitude** (height of the wave)
- **Frequency** (how fast it oscillates)
- **Phase** (where it starts in its cycle)

All analog and digital communication systems use these three building blocks.



Modulation	What Changes	Advantages	Disadvantages
AM (Amplitude Modulation)	Carrier amplitude varies	+ Very simple to generate and demodulate + Cheap receivers + Long-range	= Very sensitive to noise (noise = amplitude changes) = Poor SNR = Inefficient use of power & bandwidth
FM (Frequency Modulation)	Carrier frequency varies	+ Great noise immunity + High audio quality + Used in radio, voice, telemetry	= Requires more bandwidth = More complex receivers
PM (Phase Modulation)	Carrier phase varies	+ Good noise immunity + Basis of modern digital modulation (PSK/QAM)	= Requires more complex demodulation = Phase distortions cause symbol errors

Modulation	What Changes	Advantages	Disadvantages
ASK (Amplitude Shift Keying)	Amplitude switches between levels	+ Very simple + Low cost	= Very susceptible to noise = Low data rate
FSK (Frequency Shift Keying)	Frequency switches between tones	+ Very robust + Good for low-SNR links (LoRa-like)	= Low spectral efficiency = Requires larger bandwidth
PSK (Phase Shift Keying)	Phase jumps between discrete angles	+ Very power-efficient + Good spectral efficiency	= Requires accurate phase recovery = Sensitive to phase noise
QPSK (Quadrature PSK)	4 distinct phase states → 2 bits/symbol	+ Doubles bit rate of BPSK + Widely used (Wi-Fi, LTE) + Good balance of complexity vs speed	= Still sensitive to phase noise = Requires complex demodulator
QAM (16, 64, 256, 1024...)	Amplitude + phase simultaneously	+ Extremely high data rates + Excellent spectral efficiency + Backbone of Wi-Fi, LTE, 5G	= Requires high SNR = Very sensitive to distortion = Needs linear PA & precise LO

QAM Modulation

The modulation that gives Wi-Fi, LTE, and 5G their high speed.

QAM changes BOTH the amplitude and the phase of the carrier at the same time.

- This allows each symbol to represent **multiple bits**, instead of just 1 bit (ASK/FSK) or 2 bits (QPSK).

Think of QAM as sending information on:

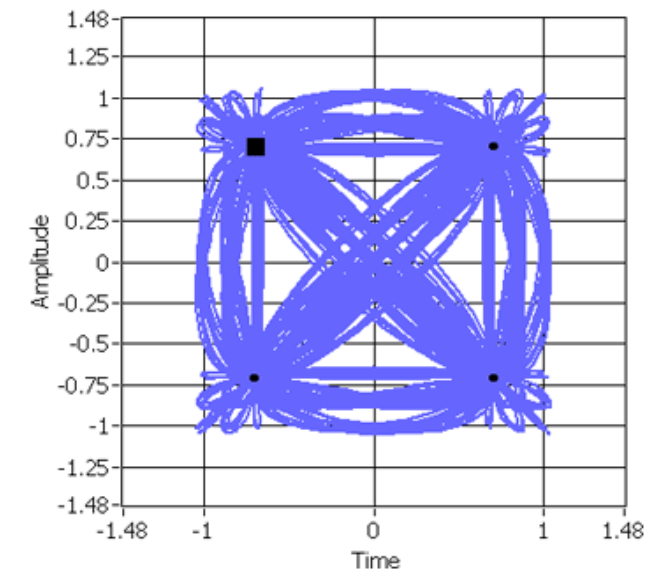
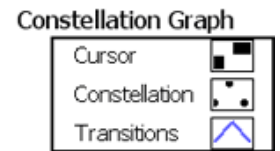
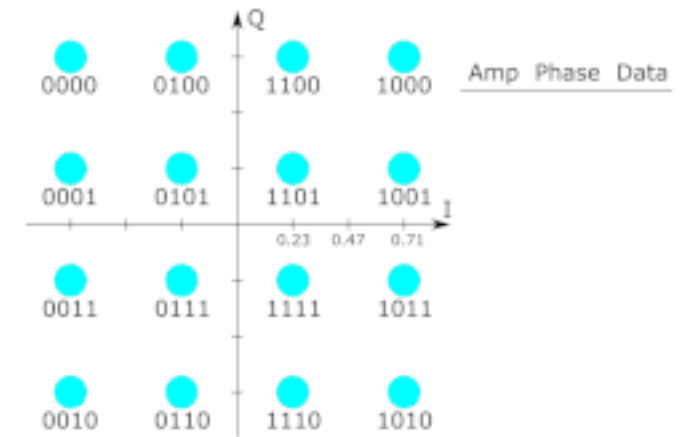
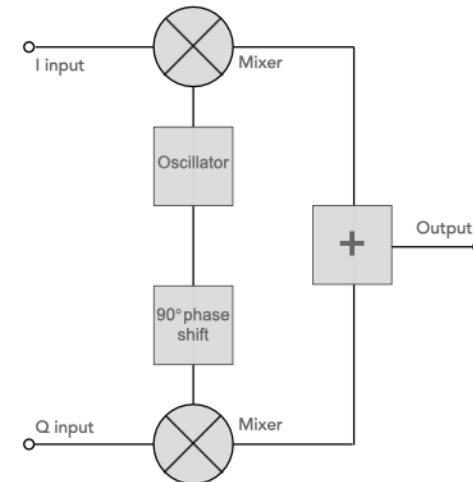
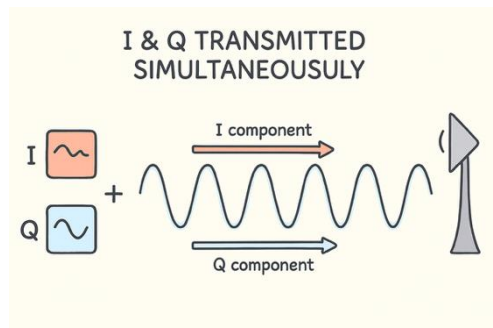
- I (In-phase) axis → amplitude
- Q (Quadrature) axis → phase shift

Each point on the I/Q plane = one “symbol”. We actually can send 2 values at the same time on the same carrier because they are 90 degrees apart (orthogonal). **Each SYMBOL in QAM is actually 2 values sent at once.**

QAM is Extremely high spectral efficiency. Also we don't send bits we send symbols with multiple bits.

Mathematically The transmitted wave is:

$$s(t) = I \cdot \cos(\omega t) - Q \cdot \sin(\omega t)$$



QAM Demodulation

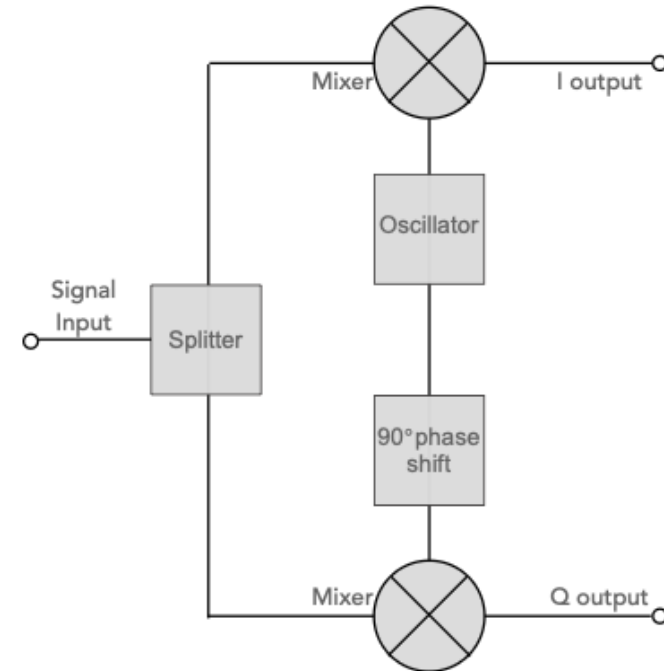
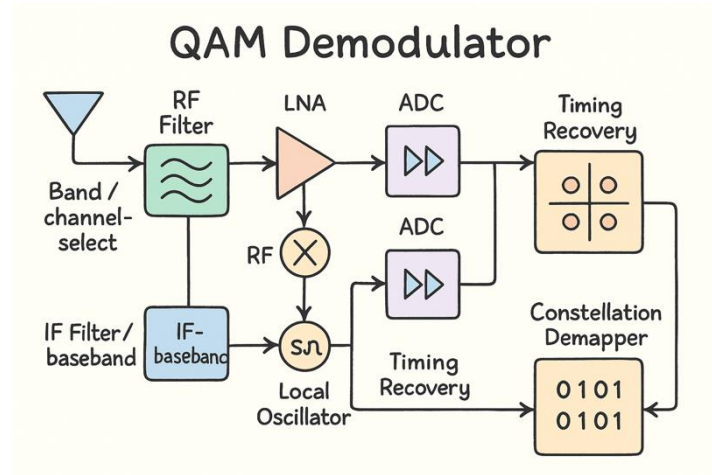
To demodulate QAM, the receiver must **recover the I and Q values** that were transmitted *simultaneously* on the same carrier.

The trick:

- We use **two local oscillators** 90° apart (cosine & sine) to “separate” the I and Q pieces.

To demodulate QAM:

- **Multiply** the received signal by a cosine → extracts **I**
- **Multiply** the received signal by a sine → extracts **Q**
- **Low-pass filter** both → isolate baseband
- **Sample + detect** the closest point → recover bits



OFDM

OFDM sends data using **many narrow subcarriers at the same time**, instead of one wide carrier.

Each subcarrier carries a **QAM symbol**, and all subcarriers are spaced so precisely that they are **orthogonal** (no interference).



How It Works

- Split the data stream into many slower streams
- Assign each stream to a **different subcarrier**
- Modulate each subcarrier with QPSK/QAM
- Use **IFFT** to combine all subcarriers into one time-domain signal
- Transmit
- At the receiver, use **FFT** to separate the subcarriers again



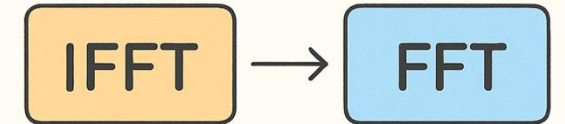
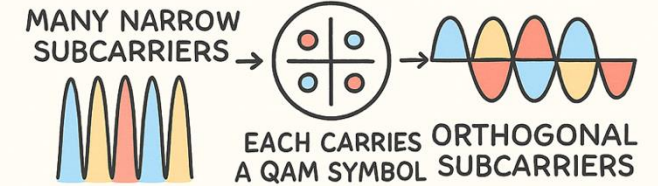
Used In

- Wi-Fi (802.11a/g/n/ac/ax/be)
- LTE / 4G
- 5G NR
- Digital TV (DVB-T)

This way we can send data in parallel without overloading the channels.

OFDM

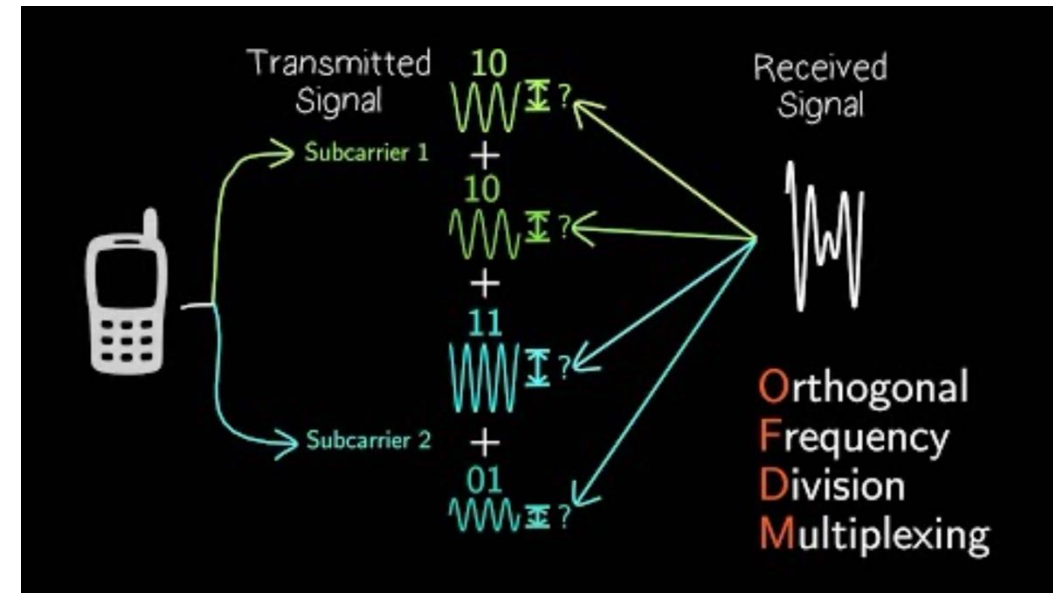
ORTHOGONAL FREQUENCY DIVISION MULTIPLEXING



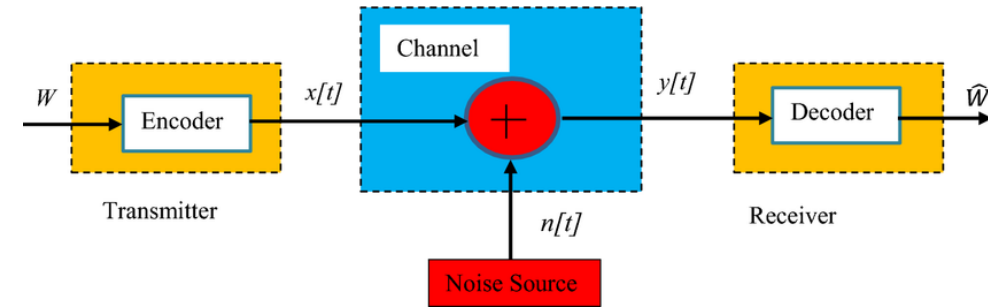
OFDM splits the channel into many narrow, orthogonal subcarriers.

Each carries a QAM symbol.

IFFT makes them; FFT separates them.



Shannon Capacity



Shannon Capacity tells us the **maximum possible data rate** that can be sent over a communication channel **without errors**, given:

- The **bandwidth** of the channel
- The **signal-to-noise ratio (SNR)**

It is the **speed limit of information theory** — no technology can beat it.

The Shannon Capacity Formula

$$C = B \log_2(1 + SNR)$$

Where:

- **C** = maximum data rate (bits/sec)
- **B** = channel bandwidth (Hz)
- **SNR** = signal-to-noise ratio (linear, not dB)

What It Means Physically

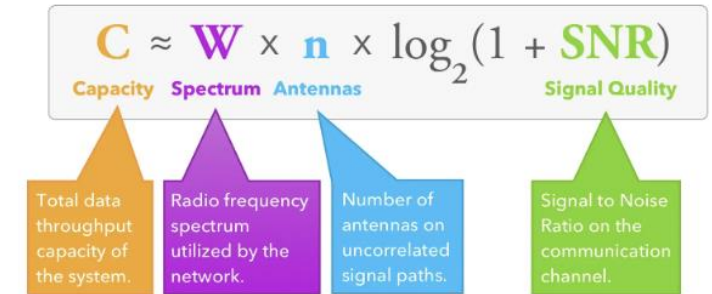
- ✓ **More bandwidth** → **more data** More “space” to send symbols.
- ✓ **More SNR** → **more bits per symbol**

Higher SNR allows using higher-order QAM (16-QAM, 64-QAM, 256-QAM, etc). For example

20 MHz Wi-Fi channel, SNR = 20 dB → SNR ≈ 100 linear

$$C = 20\text{Mhz} \log_2(1 + 100)$$

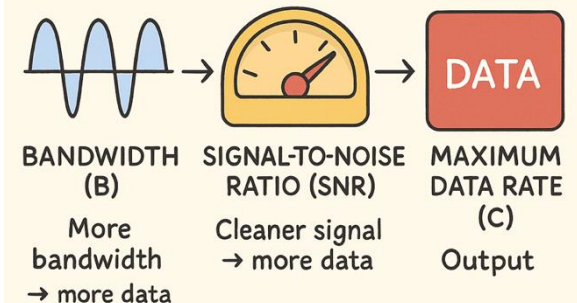
$$C \approx 20\text{MHz} \cdot 6.66 \approx 133 \text{ Mbps}$$



SHANNON CAPACITY

$$C = B \log_2(1 + SNR)$$

The maximum data rate of a communication channel



Superheterodyne Receiver

The Most Successful Receiver Architecture in RF History

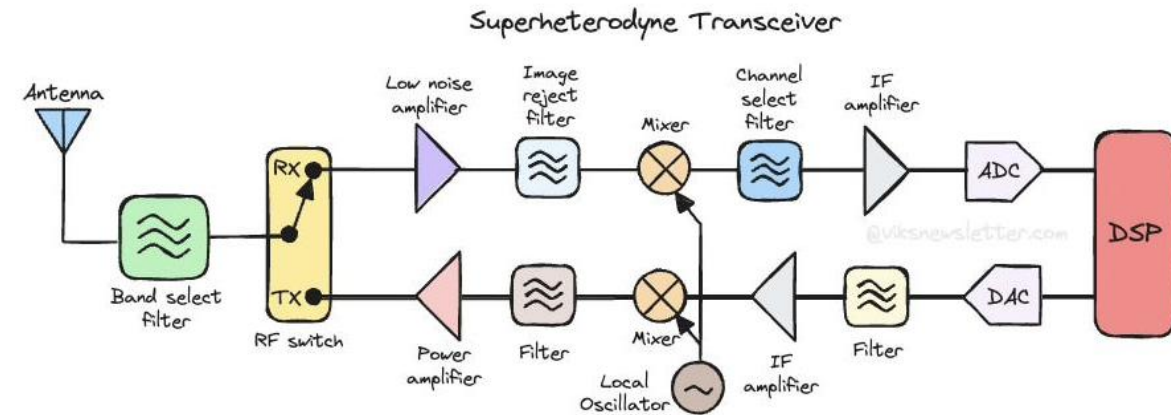
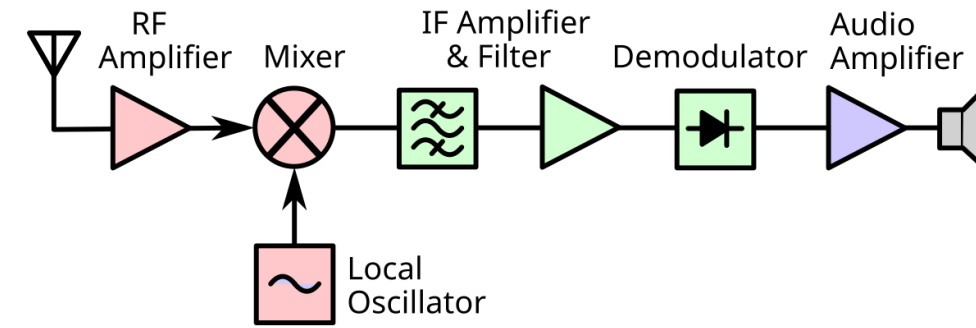
Directly filtering or demodulating a high-frequency RF signal is difficult. So instead, the superheterodyne receiver **converts** the RF signal to a lower, easier-to-handle frequency called the **IF (Intermediate Frequency)**.

Why Superheterodyne Was Revolutionary

- Allowed **long-distance radio** reception
- Enabled **narrow channels** and **excellent filtering**
- Standard for:
 - AM/FM radios
 - TV tuners
 - Radar receivers
 - Shortwave radios
 - Military radios
- Many SDRs today

Invented by Edwin Armstrong (1918), still used in modern RF systems.

We can also use the same idea for transmit data.



RF Filters

RF filters allow some frequencies to pass and block others. They shape the spectrum so the radio only sees what it needs.

Types:

LPF (Low-Pass Filter)

- Passes low frequencies, blocks high.
- Used after mixers, in baseband, or to reduce harmonics.

BPF (Band-Pass Filter)

- Allows one frequency band through, rejects others.
- Used on the antenna path (front-end) and before LNAs/PAs.

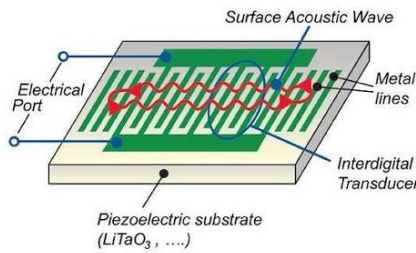
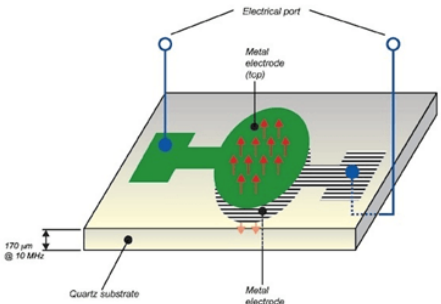
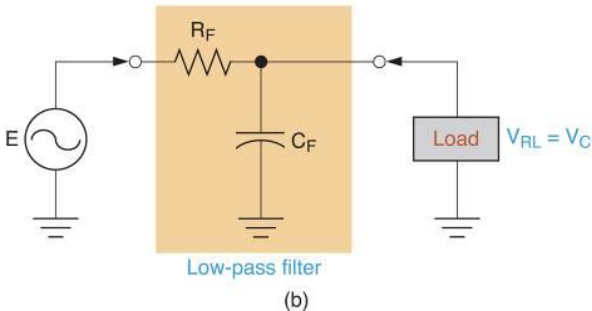
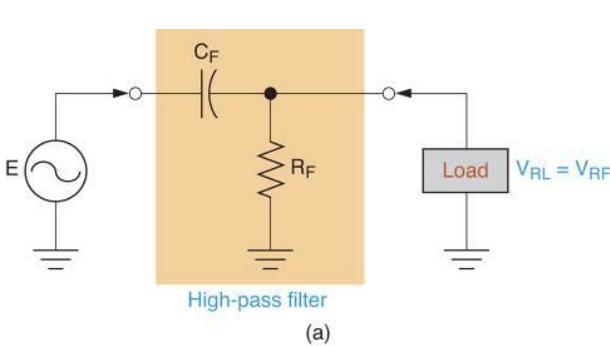
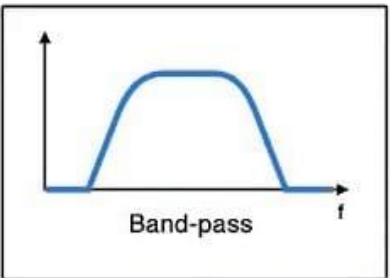
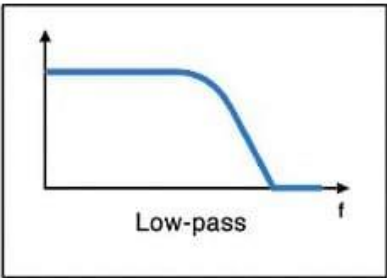
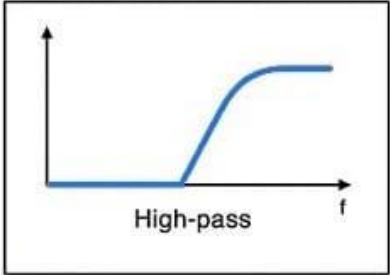
SAW & BAW Filters (Used in Phones)

- Built using crystal-like acoustic resonators.
- Extremely sharp cutoff (very high Q-factor).
- Essential for LTE/5G/Wi-Fi band filtering and duplexers.

LC Filters (Inductor + Capacitor)

- Used for low-frequency RF (<500 MHz), IF filters, matching networks, and tunable filters.
- Lower Q-factor, so not sharp enough for tightly packed GHz bands.
- Still very common in low-cost radios and baseband filtering.

Low Pass,
High Pass
and Band
Pass Filters



Q-Factor & Cutoff Sharpness

† What Is Q-Factor?

The **Quality Factor (Q)** tells how “selective” or “sharp” a resonant circuit or filter is.

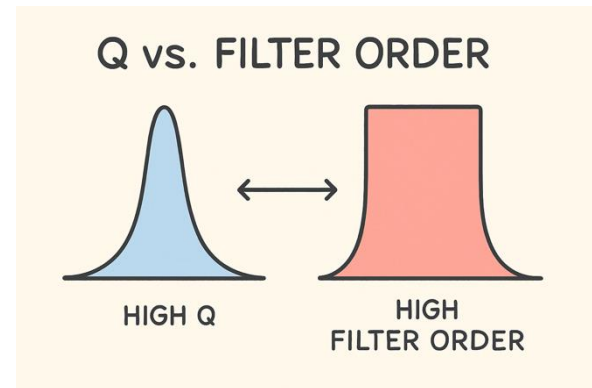
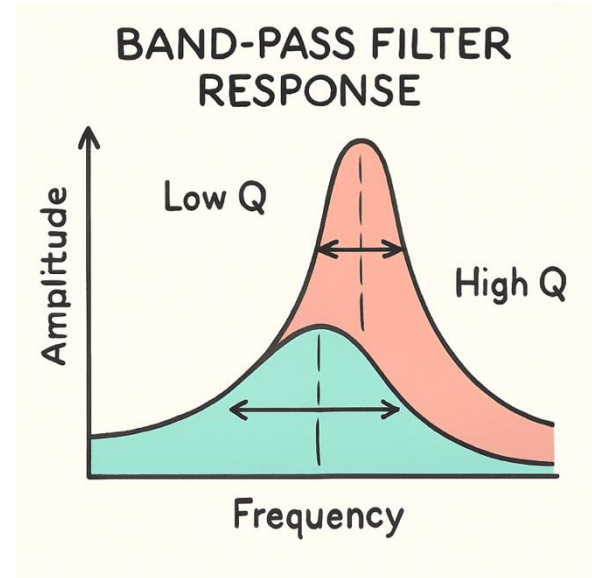
$$Q = \frac{f_0}{\text{Bandwidth}}$$

- **High Q** → narrow bandwidth → very selective
- **Low Q** → wide bandwidth → gentle filtering

While the Q-factor defines how “tall and skinny” the mountain is, the Filter order defines how “steep the cliffs” are

† What “Order” Means

The **order** of a filter refers to the number of **energy-storing elements** (inductors, capacitors, resonators, poles) in the filter.



Impedance Matching

Impedance matching means making every block in the RF chain look like the same impedance (usually 50 Ω). When impedances don't match, part of the signal bounces back instead of being delivered to the next stage.

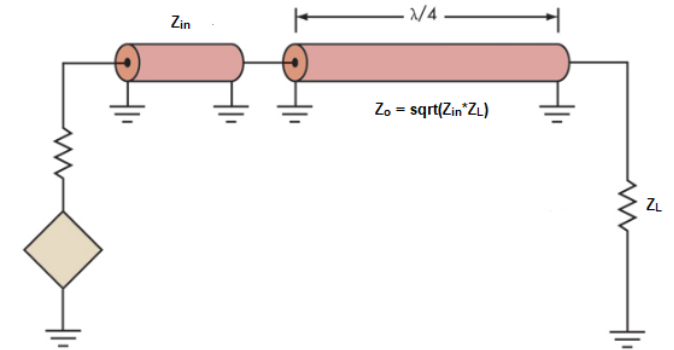
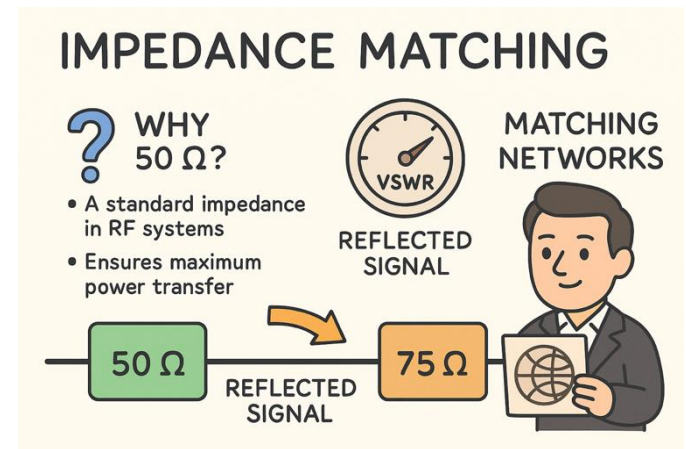
Why 50 Ω ?

- A compromise between maximum power transfer and minimum loss.
- Standard for RF cables, antennas, LNAs, PAs, SAW/BAW filters.
- Makes RF systems compatible and predictable.

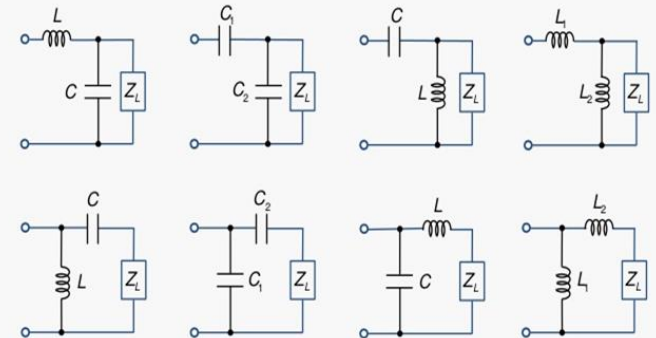
We need to design Matching Networks in between components to achieve impedance matching.

Matching Networks (L, Pi, T)

- Use inductors & capacitors to transform impedances.
- Bring the load back to 50 Ω at the operating frequency.
- Used between antenna $\leftrightarrow$ tuner $\leftrightarrow$ filter $\leftrightarrow$ LNA/PA.
- Smith Chart shows how each component moves impedance.



Matching Networks (Two-Element L-Shape)



Smith Chart

The Smith Chart is a graphical tool that shows how impedances behave in RF

- circuits. It helps engineers design matching networks by visualizing how
- inductors and capacitors move impedance along the chart.

Why It Exists

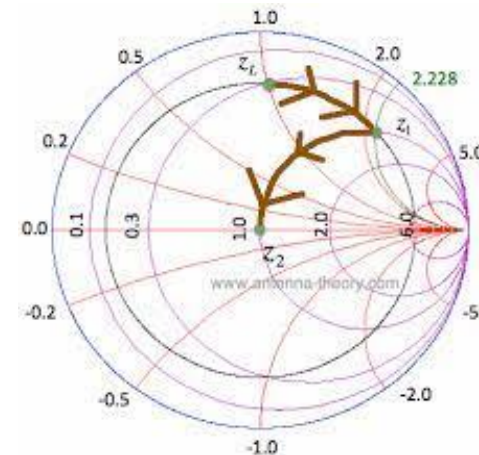
- At RF, impedance is complex ($R + jX$), not just resistance.
- Transmission lines and reactive components transform impedance in non-intuitive ways.

The Smith Chart makes these transformations visual and simple.

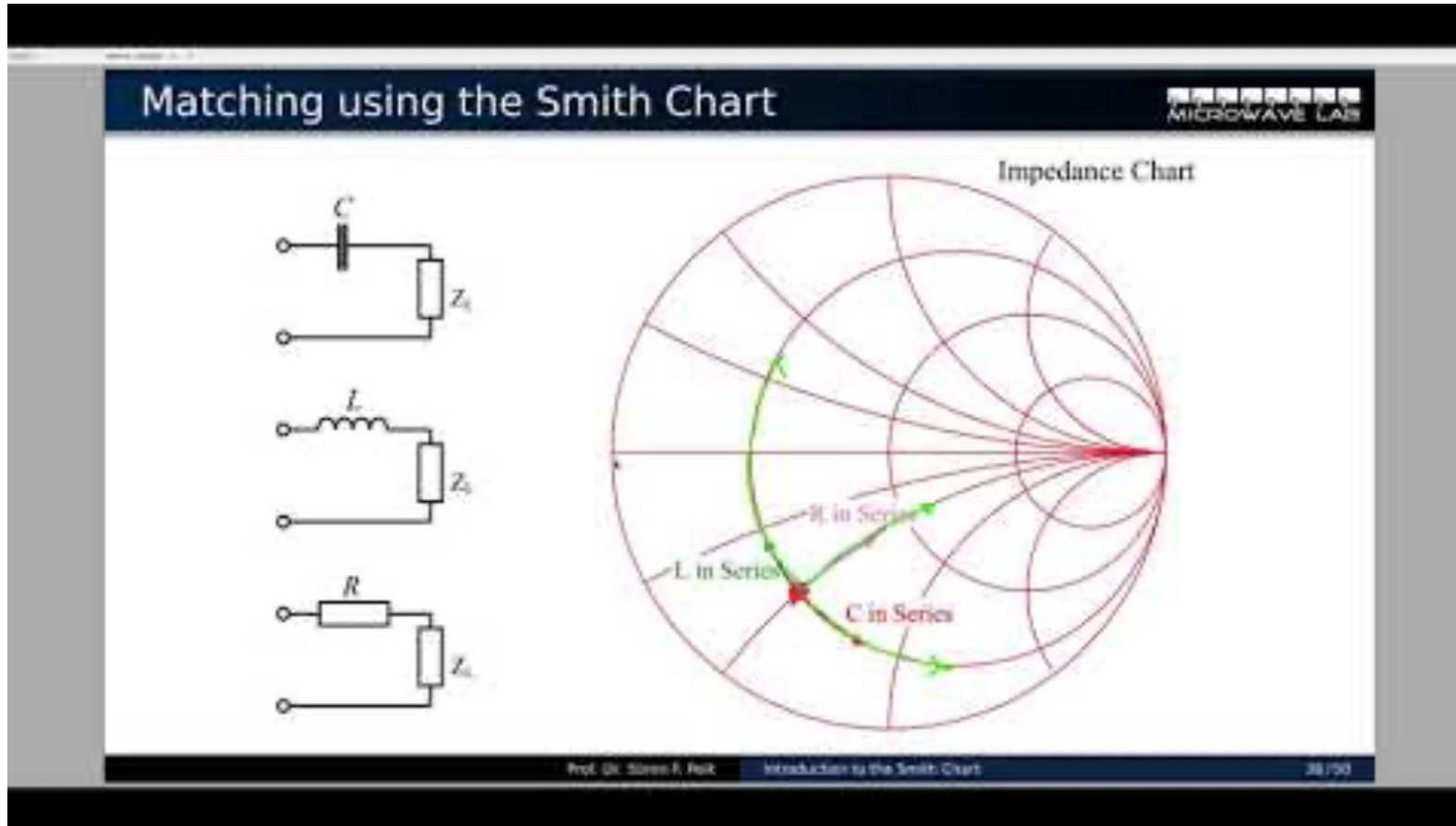
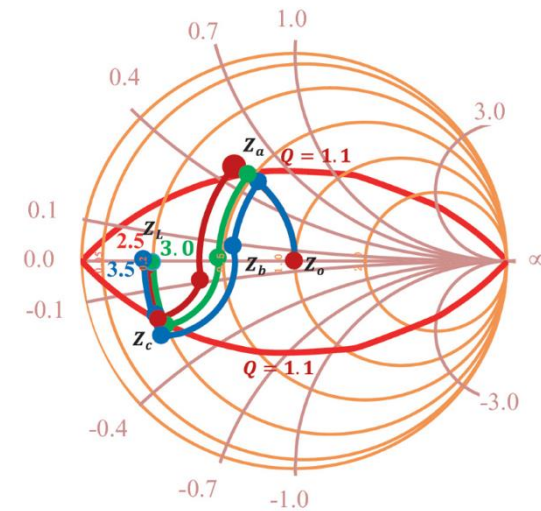
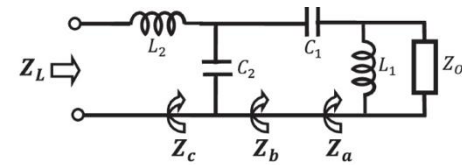
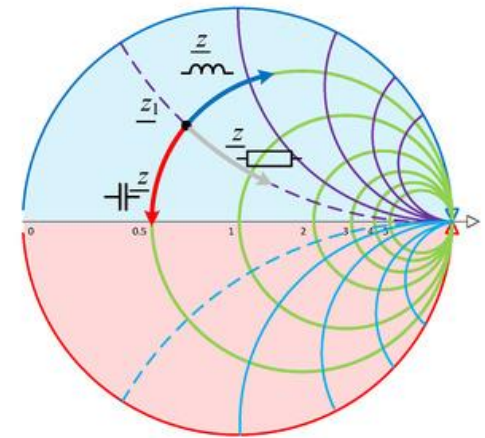
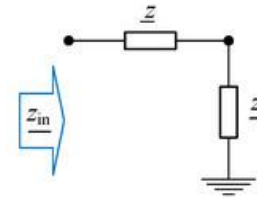
Resistors, Capacitors or Inductors, in parallel (shunt) or series move into the Smith chart, in different directions, the designer job is to make the dots connect.

How It Works

- The center of the chart = perfect match ($50\ \Omega$).
- Right side = high resistance; left side = low resistance.
- Upper half = inductive (positive reactance).
- Lower half = capacitive (negative reactance).
- Circles represent constant resistance.
- Arcs represent constant reactance.



Smith Chart Matching



Practical Matching Example

Match a complex load $Z_L = 25 + j50 \Omega$ to a 50Ω system using an L-network.

1) Normalize the load

- $Z_0 = 50 \Omega$
- $z_L = Z_L / Z_0 = (25 + j50) / 50 = 0.5 + j1$

On the Smith Chart: point at $r = 0.5$, $x = +1$ (inductive).

2) Add a series reactance (X_s)

Choose a series capacitor so that the admittance of the new point has the correct conductance ($1/50 \text{ S}$).

Solving (or using the Smith Chart) gives two valid solutions; we pick:

- $X_s = -25 \Omega$ (series capacitor)
- New impedance:
- $Z_1 = 25 + j(50 - 25) = 25 + j25 \Omega$

3) Convert to admittance $Y_1 = 1 / Z_1 \approx 0.02 - j0.02 \text{ S}$

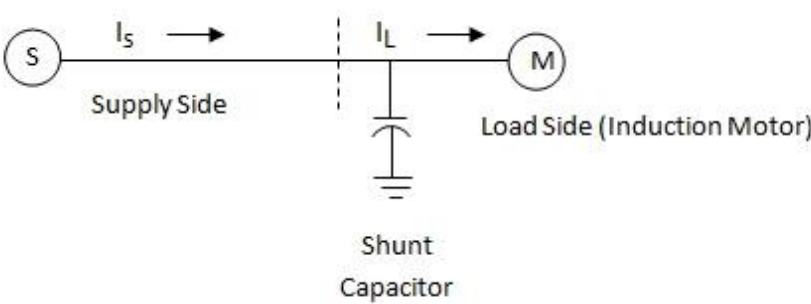
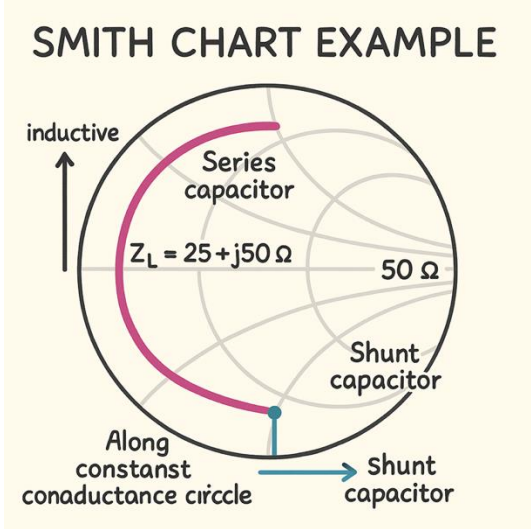
- Conductance $G_1 = 0.02 \text{ S}$ (good $\rightarrow$ equals $1/50$)
- Susceptance $B_1 = -0.02 \text{ S}$ (inductive)

On the Smith Chart: we are on the 50Ω conductance circle, but still below the real axis (inductive).

4) Add a shunt susceptance (B_p) to cancel B_1 $B_p = +0.02 \text{ S}$ (capacitive shunt)

Total susceptance:

$B_{\text{total}} = B_1 + B_p = -0.02 + 0.02 = 0$
 $\rightarrow Y_{\text{total}} = 0.02 + j0 = 1 / 50 \text{ S}$
 $\rightarrow Z_{\text{total}} = 50 \Omega$ (perfect match)



	Shunt	Series
Basic schematic		
Strengths	- No maximum VIN - Accommodates large VIN ranges with proper Rs - Can be used as a negative reference - Can be used as floating reference (cathode not grounded)	- Lower power dissipation - No need for external resistor - Low Dropout - Can have extra pins with additional features
Use cases	- Wide VIN - VIN transients - Floating/negative references	- Load current variation - Low-power consumption requirements

Table 1: Typical comparison table of shunt versus series voltage references

Transmission Lines

At low frequencies, wires behave like simple connections. At RF, wires become “transmission lines,” where voltage and current travel as electromagnetic waves along the conductor.

When the physical length of a trace or cable becomes larger than $\sim \lambda/10$, the line has distributed inductance and capacitance:

- It stores energy
- It delays signals
- It reflects power if impedance mismatches occur

Transmission Line Rules:

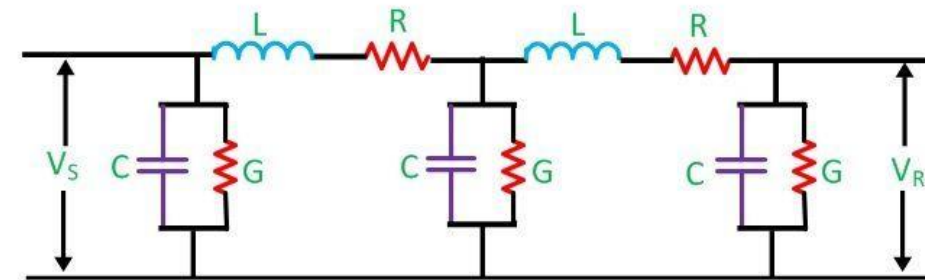
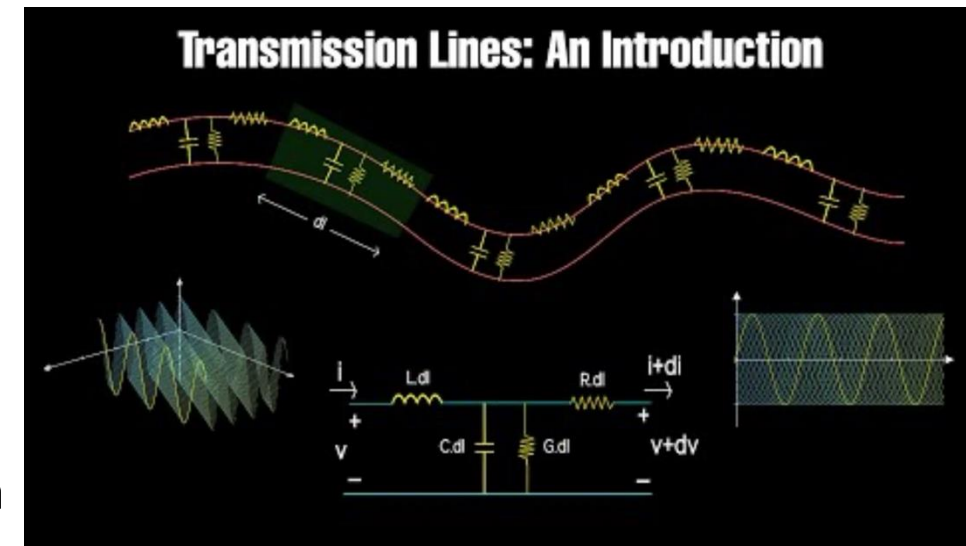
- Coax, microstrip, CPW, twisted pair $\rightarrow$ all are transmission lines.
- They each have a characteristic impedance (Z_0), typically 50 Ω in RF.

Goal of RF Design:

Keep the whole RF chain at 50 Ω to prevent reflections and maintain maximum power flow from antenna $\rightarrow$ filters $\rightarrow$ mixers $\rightarrow$ amplifiers.

Core Idea:

- At RF, your PCB traces are no longer wires — they are waveguides.

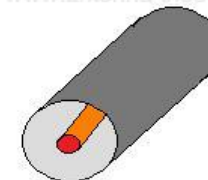


Transmission Line Model

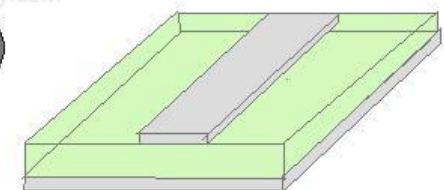
$$Z = R + j\omega L, Y = G + j\omega C$$

Circuit Globe

www.antenna-theory.com

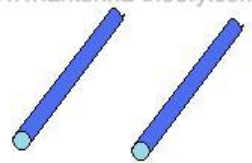


Coaxial Cable



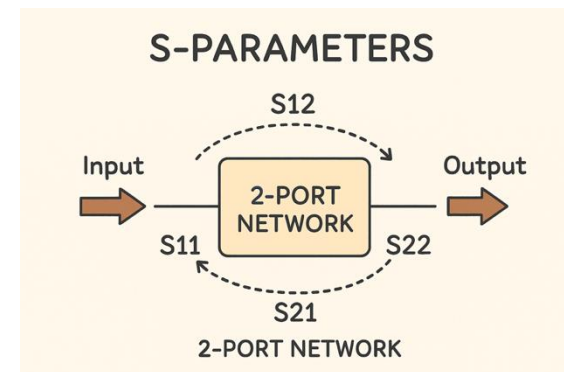
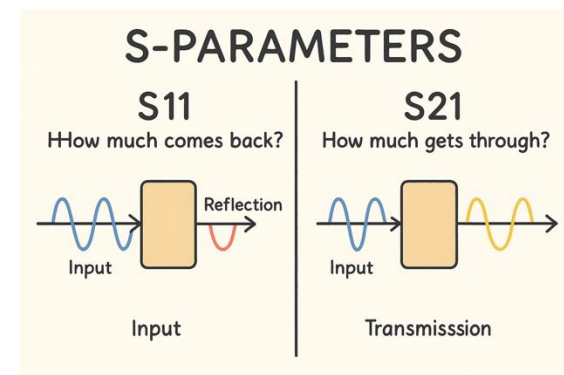
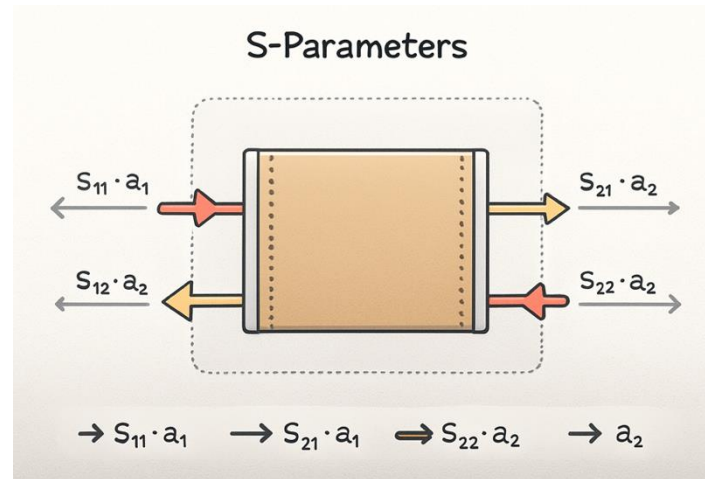
Microstrip Line

www.antenna-theory.com



Two-Wire Line

S-Parameter



S-parameters (Scattering Parameters) describe how RF signals behave when they hit a circuit. Instead of measuring voltage and current (hard at GHz), we measure how much of the wave is transmitted or reflected.

Each S-parameter represents a ratio of waves:

- S_{11} = Reflection (input) “How much comes back?”
- S_{21} = Forward transmission (gain/loss) “How much gets through?”
- S_{12} = Reverse isolation (leakage backwards) “How much signal leaks backward.”
- S_{22} = Output reflection “How much comes back on the output?”

Why S-Parameters Matter:

- They define how real RF components behave: antennas, filters, LNAs, mixers, PAs.
- They are what VNAs measure.
- Let you design matching networks using Smith Chart.
- Frequency-dependent — you see their behavior across the RF band.

What are S-parameters?

The diagram shows a person pointing to a network box labeled "The Network" with "Port 1" and "Port 2". At Port 1, an incident wave V_1^+ and a reflected wave V_1^- are shown. At Port 2, a transmitted wave V_2^- is shown. The S-parameters S_{11} and S_{21} are indicated.

Voltage at Port 1: $V_1 = V_1^+ + V_1^- = V_1^+(1 + S_{11})$
 Voltage at Port 2: $V_2 = V_2^- = V_1^+ S_{21}$

LNA (Low Noise Amplifiers)

The LNA (Low Noise Amplifier) is the first active component in the RF receiver. Its job is to amplify very weak signals coming from the antenna while adding as little noise and distortion as possible.

Why the LNA Is Critical:

- Signals arriving at the antenna are extremely small (-90 to -120 dBm).
- Any noise added at this stage cannot be removed later.
- The LNA sets the overall receiver sensitivity and noise figure.

Key Requirements of an LNA:

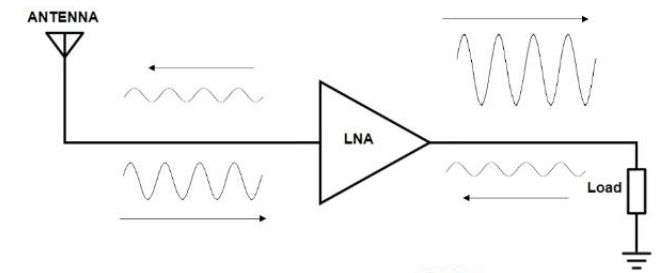
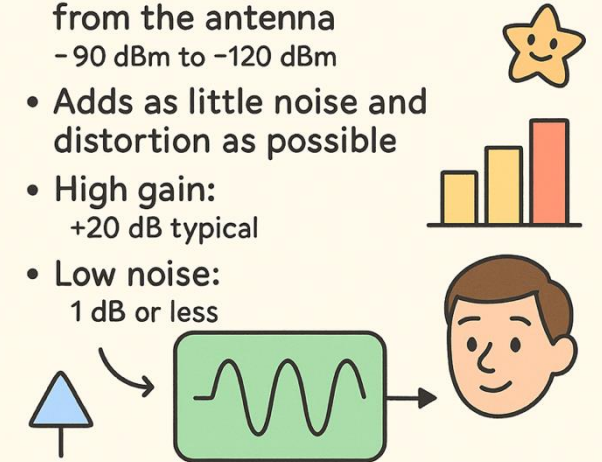
- Low Noise Figure (NF) and High Gain
- Good Linearity
- 50 ohms input impedance
- No Oscillation

What the LNA Does NOT Do:

- It does not filter the signal — that's the filter's job.
- It does not convert frequency — that's the mixer's job.

LNA – Low Noise Amplifier

- Amplifies very weak signals from the antenna
 -90 dBm to -120 dBm
- Adds as little noise and distortion as possible
- High gain:
 $+20$ dB typical
- Low noise:
 1 dB or less



Mixers, Upconverters & Downconverters

A **mixer** is an RF component that takes **two signals** and produces **new frequencies** equal to the *sum and difference* of their frequencies. If you feed:

- Signal A at frequency f_1
- Local Oscillator (LO) at frequency f_2

The mixer outputs: **$(f_1 + f_2)$ and $(f_1 - f_2)$**

This is the whole magic of frequency translation.

- It's used for:
- moving RF down to IF (**downconversion**)
- moving IF up to RF (**upconversion**)
- modulation
- channelization
- detection
- software-defined radios

A mixer is essentially a **multiplication device**:

$$out(t) = f_1(t) \times \cos(2\pi \cdot f_{LO}t)$$

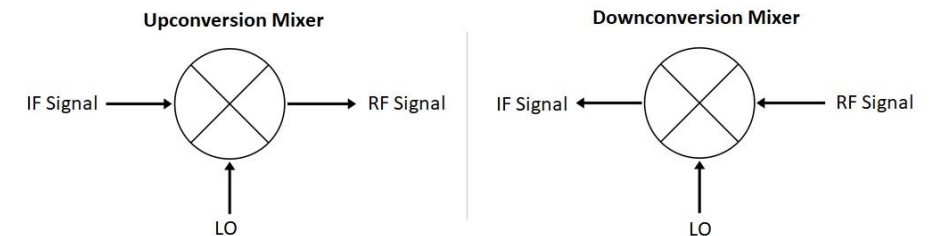
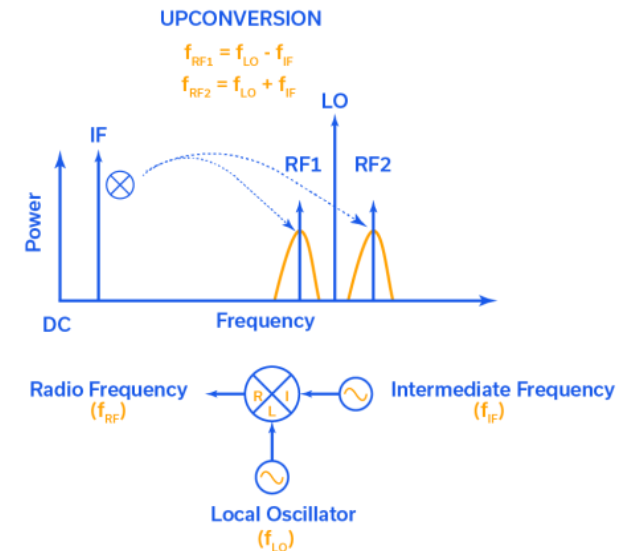
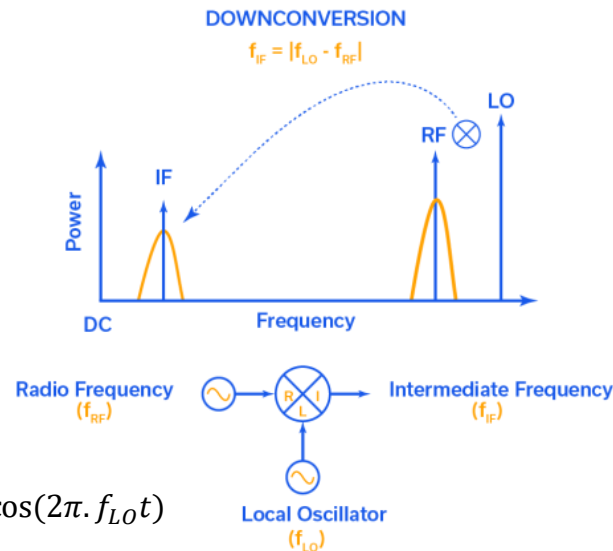
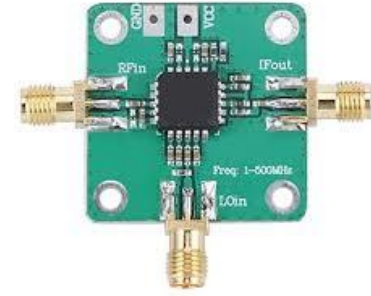
Why Downconvert?

Because designing **filters**, **ADCs**, **amps** at RF is extremely hard. At IF (MHz range), everything becomes MUCH easier.

Why Upconvert?

Because transmitters must operate at:

- Wi-Fi frequencies
- Cellular bands
- Radar bands
- RF spectrum allocated by regulators



Intermediate Frequency

An **Intermediate Frequency** is a *middle frequency* created inside a radio receiver (or transmitter) after mixing the signal with a **Local Oscillator (LO)**.

- It is **not** the antenna frequency (RF)
- and **not** the final audio/data frequency (baseband).
- It is something **in between**.



Where IF comes from:

A receiver takes the incoming RF signal and mixes it with the LO:

$$RF \pm LO \rightarrow IF$$

Mixing “shifts” the signal to a new frequency, the **intermediate frequency**, where:

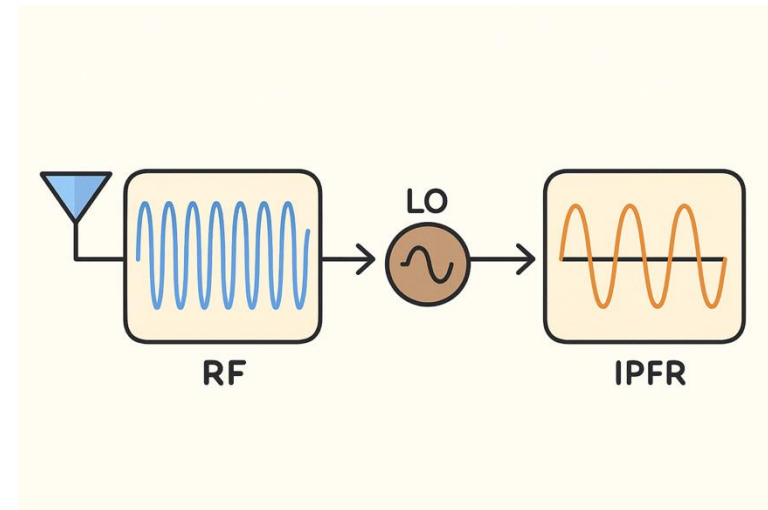
- filters work better
- amplifiers are more stable
- demodulation becomes easier

It's basically a Frequency Where is easier to compute / operate.

Before digital RF existed, it was difficult to build:

- super narrow filters at high GHz
- low-noise amplifiers at any frequency
- stable oscillators at all bands

So engineers invented the **superheterodyne receiver** (1920s)



RF Oscillators & Local Oscillators (LO)

The Heartbeat of Every Wireless System

An oscillator is a circuit that generates a **continuous AC signal** at a stable frequency.

In RF systems, oscillators are used to create **sine waves**, **carriers**, and **clock signals**. They turn **DC power** → a **precise RF tone**.

What Is an LO (Local Oscillator)?

A **Local Oscillator** is a stable RF signal used inside a receiver or transmitter to **mix** signals and change frequency.

In a receiver:

- LO + incoming RF → produces the **Intermediate Frequency (IF)**
- (heterodyne architecture)

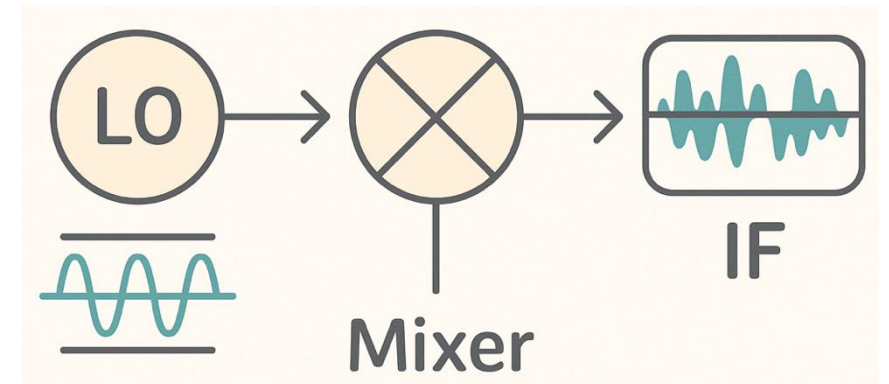
In a transmitter:

- Baseband + LO → upconverts the signal to the RF carrier.
- **Every radio, Wi-Fi, 5G phone, and radar uses an LO.**

Why Oscillators Matter

- Define the **carrier frequency**
- Allow **mixing / upconversion / downconversion**
- Set the **channel spacing**
- Control **modulation accuracy**
- Determine **spectral purity** (phase noise, jitter)

Oscillators are the **frequency reference** of all RF systems.



PLL & Frequency Synthesis — How Radios Create Precise Frequencies

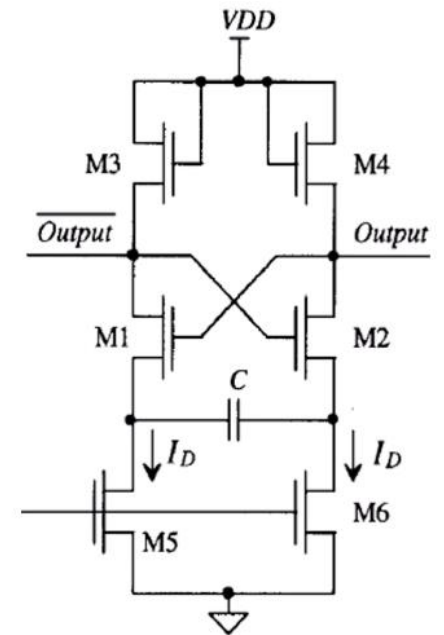
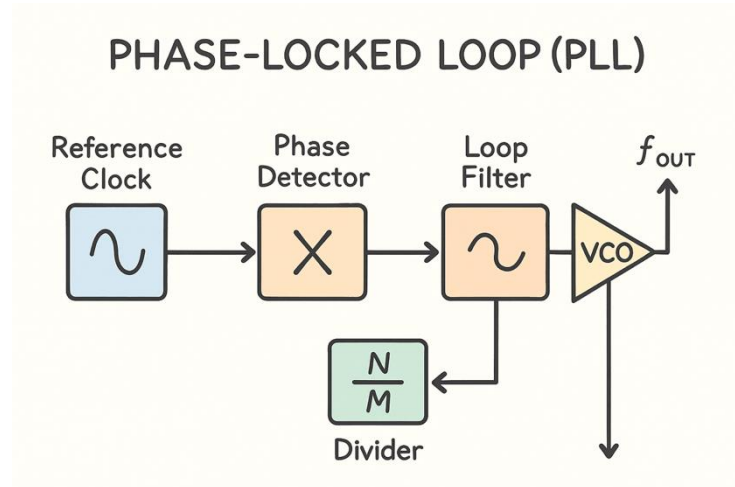
Radios need **very accurate and tunable frequencies** for:

- Local Oscillator (LO)
- Mixers (up/down conversion)
- Channel selection
- OFDM subcarriers

But a crystal oscillator (ie 40Mhz) can only produce **one fixed frequency**. A PLL solves this by creating **any frequency you want**, locked to a stable reference.

A **Phase-Locked Loop (PLL)** is a control system that:

- Compares the **phase** of two signals
- Adjusts a **VCO (voltage-controlled oscillator)**
- Forces the VCO to match the reference frequency (or a multiple of it)



Noise, SNR & Noise Figure

Noise is **unwanted random energy** that adds to your signal and makes it harder to decode. In RF systems, major noise sources are:

- **Thermal noise** (the fundamental physical limit)
- **Receiver electronics** (LNA, mixers, ADC)
- **Interference** from other devices

Wider bandwidth = more noise. Thermal noise is unavoidable.



What Is SNR (Signal-to-Noise Ratio)?

SNR determines how close you can get to **Shannon's capacity limit**.

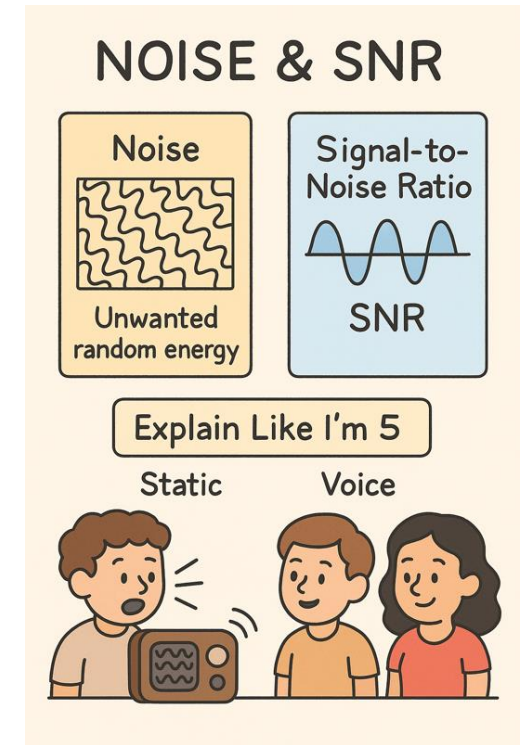


What Is Noise Figure (NF)?

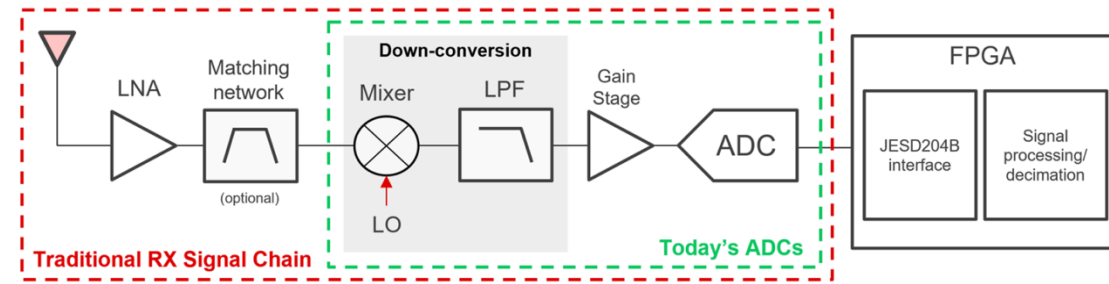
NF determines how faint a signal the receiver can detect.

In simpler words

- Noise = static
- SNR = how loud your friend is vs the static
- Noise figure = how much extra static your **radio** adds



RF Frontend



The RF front-end is the section of the radio that takes an electromagnetic wave from the antenna and converts it into clean baseband samples for the DSP processor. Each block performs a specific function to improve signal quality.

1) Antenna

- Captures the RF signal from the air.
- Must be matched to $50\ \Omega$ to avoid reflections.

2) RF Filter (SAW/BAW or LC)

- Selects the desired band.
- Rejects strong out-of-band blockers.
- Protects the LNA from overload.

3) LNA — Low Noise Amplifier

- Amplifies very weak signals without adding noise.
- Sets the receiver's sensitivity (NF).

4) Mixer + Local Oscillator (Downconversion)

- Shifts the signal from RF to a lower frequency (IF or zero-IF).
- Enables easier filtering and amplification.
- Outputs I/Q components in modern receivers.

5) IF Filter / Channel Filter

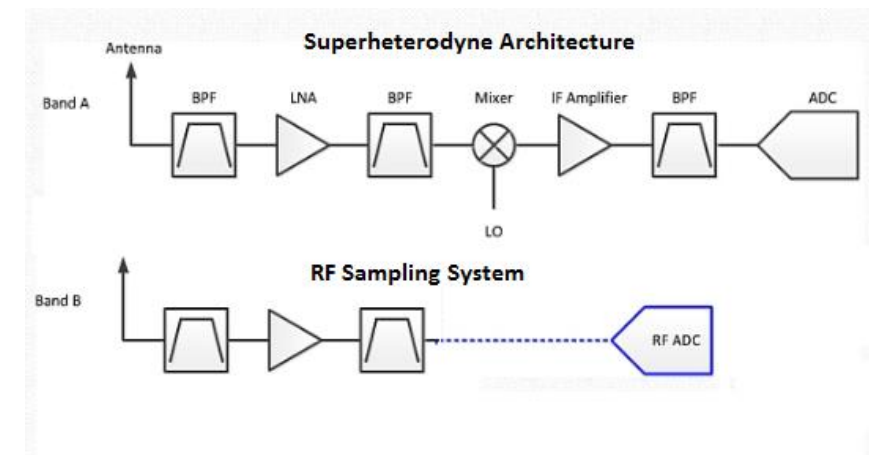
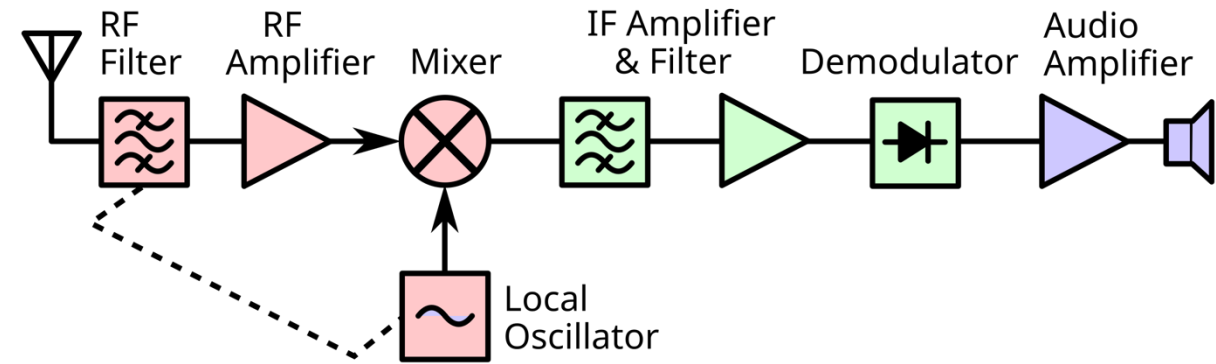
- Removes mixer images and adjacent channels.
- Defines the channel bandwidth.

6) IF Amplifier / Baseband Amplifier

- Boosts the filtered signal.
- Maintains linearity for QAM/OFDM.

7) ADC (Analog-to-Digital Converter)

- Converts I/Q analog signals into digital samples.
- Sample rate and resolution define dynamic range and SNR.
- Output: digital I/Q stream for the DSP.



Zero-IF Architecture

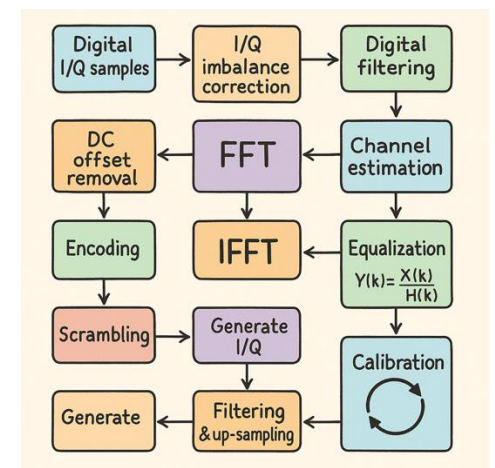
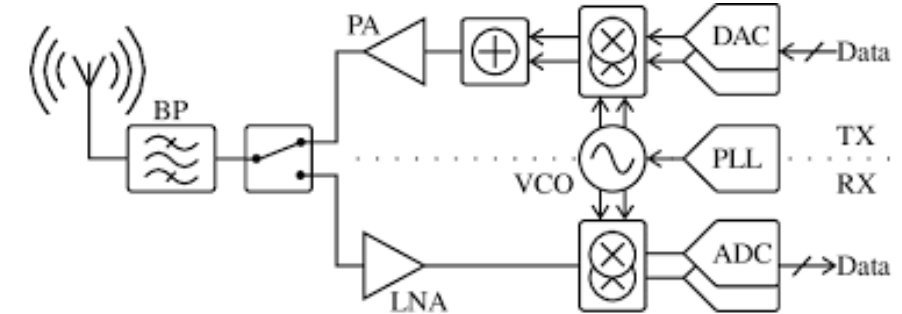
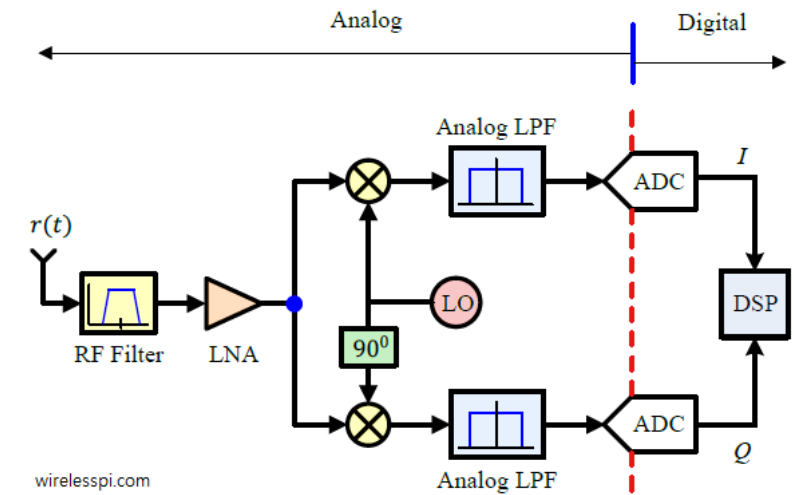
Zero-IF (also called *Direct Conversion* or *Homodyne*) is a radio architecture where the incoming RF signal is mixed directly down to 0 Hz (baseband), so there are no multiple steps like the Superheterodyne, but will need a DSP microprocessor.

The DSP processor will have to compute things like:

- FFT and iFFT for OFDM
- Digital Filters (Low-pass, decimators)
- QAM Mapping/Demapping
- Error Coding (Viterbi, LDPC, Polar codes on 5G)

The idea is to substitute hardware by Software and digital circuitry, avoiding analog filters, mixers, LO oscillators.

You will see Zero-IF on 5G, Wifi, Bluetooth.

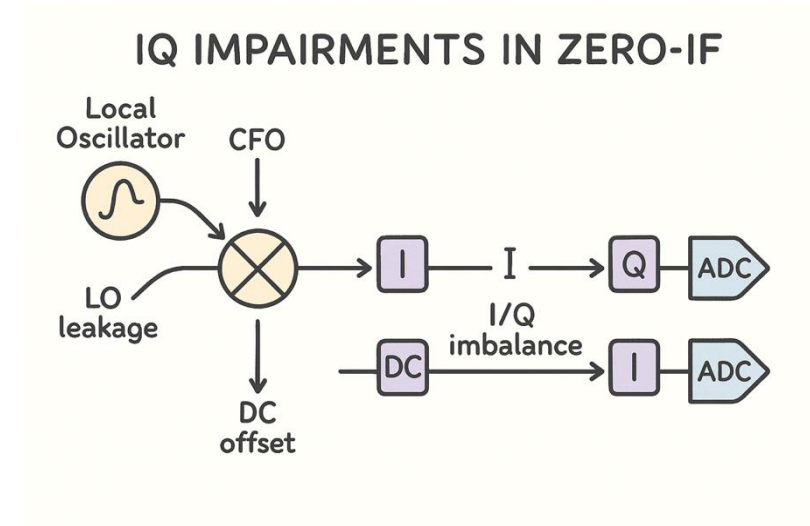


IQ Impairments — The Real-World Problems in Zero-IF Radios

When we downconvert RF signals directly to baseband (Zero-IF architecture), we create two orthogonal components: **I (In-phase)** and **Q (Quadrature)**. In theory, these should be perfect. In practice, **hardware imperfections** introduce several impairments that distort the signal.

Key Takeaway

- **In theory:** I and Q are perfect, orthogonal, and centered at DC.
- **In practice:** Hardware is imperfect. These four impairments (DC offset, I/Q imbalance, LO leakage, CFO) must be **measured, calibrated, and corrected** in every real Zero-IF radio — or the system won't work at high data rates.



Impairment	Cause	Effect	Fix
DC Offset	LO self-mixing	DC spike, ADC issues	AC coupling, DSP removal
I/Q Imbalance	Gain/phase mismatch	Image leakage, distorted QAM	DSP correction, calibration
LO Leakage	Mixer/PCB coupling	Unwanted carrier	Calibration, shielding
CFO	LO mismatch, Doppler	Phase rotation, ICI	Pilot tones, PLL correction

Power Amplifiers (PAs)

The Power Amplifier (PA) is the final active stage in the transmitter.

Its job is to boost the signal to the power level needed for the antenna, while keeping distortion low.

In modern radios (LTE/5G), the PA is one of the most critical and complex blocks.

Why PAs Matter:

- They define the transmitter's output power and range.
- They consume most of the phone's battery.
- They distort the signal if not linear (critical for QAM/OFDM).
- They must support many bands and channels.

Two Main Performance Goals:

1. Efficiency

- How much DC power is turned into RF power.
- High efficiency = less heat, longer battery life.

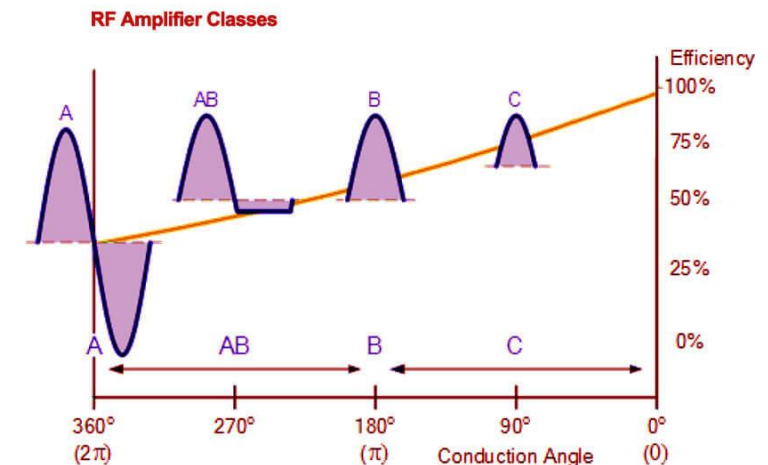
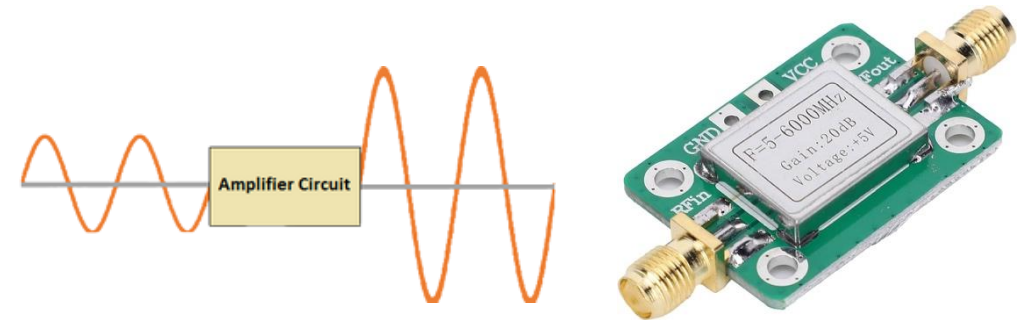
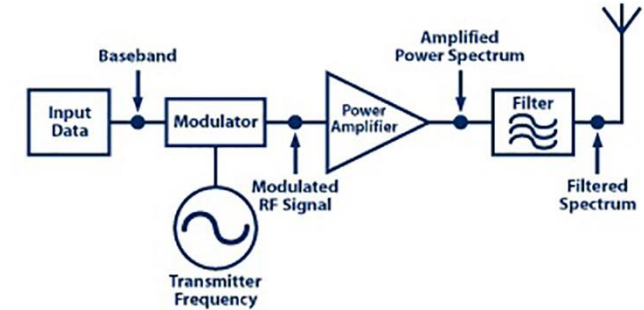
2. Linearity

- How cleanly the PA amplifies the signal without distortion.
- Critical for modulations like QAM and OFDM.

PA Efficiency Classes:

- Class A: very linear, very inefficient
- Class B/AB: good compromise (most used in phones)
- Class C/E/F: high efficiency, low linearity

The PA is the heart of the transmitter. It must be powerful, efficient, and extremely linear to support modern modulations like QAM and OFDM.



ADC/DAC Sampling

In modern radios (Zero-IF, SDR, OFDM), the ADC and DAC are the bridge between analog RF signals and the digital DSP world. Their performance limits maximum modulation order (QAM), dynamic range, and overall SNR.

Sampling & Nyquist (How fast we must sample)

An ADC takes snapshots of the signal at a sampling rate f_s .

Nyquist theorem: $f_s \geq 2 \times B$

Where B is the signal bandwidth.

If we sample too slowly:

- Aliasing occurs (frequencies fold and overlap)
- The signal becomes impossible to recover
- Every radio uses anti-alias filters before the ADC.

Quantization (How many amplitude levels we keep) (or bits)

The ADC cannot represent infinite values. It splits the voltage range into 2^n steps (n = bits).

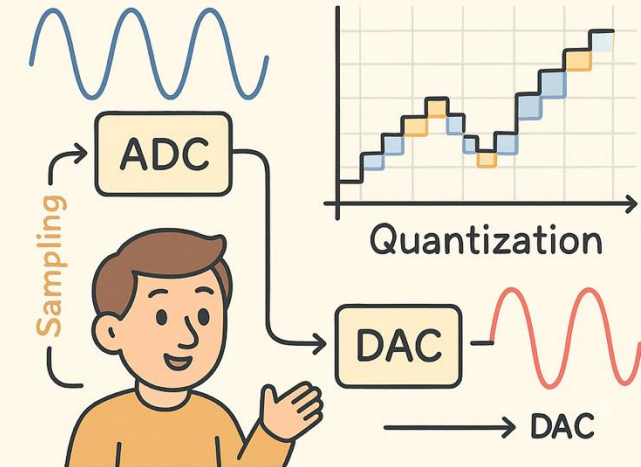
ENOB (Effective Number of Bits)

- Real ADCs are not ideal.
- Clock jitter, noise, distortion reduce effective resolution.

In Simple terms:

- ADC (Analog to Digital): We need to sample at least 2x faster than the signal
- DAC (Digital to Analog): Higher number of bit's better.

ADC & DAC SAMPLING



An ADC converts an analog signal to digital, and a DAC converts it back to analog.

MIMO and Beamforming

MIMO (Multiple-Input Multiple-Output) uses multiple antennas at the transmitter and receiver to send several data streams at the same time. Instead of one pipe, the radio now has multiple parallel pipes.

Why It Works:

- Each antenna sees a slightly different multipath environment.
- The receiver uses these differences to separate simultaneous streams.
- Increases capacity without using more bandwidth or power.

Beamforming:

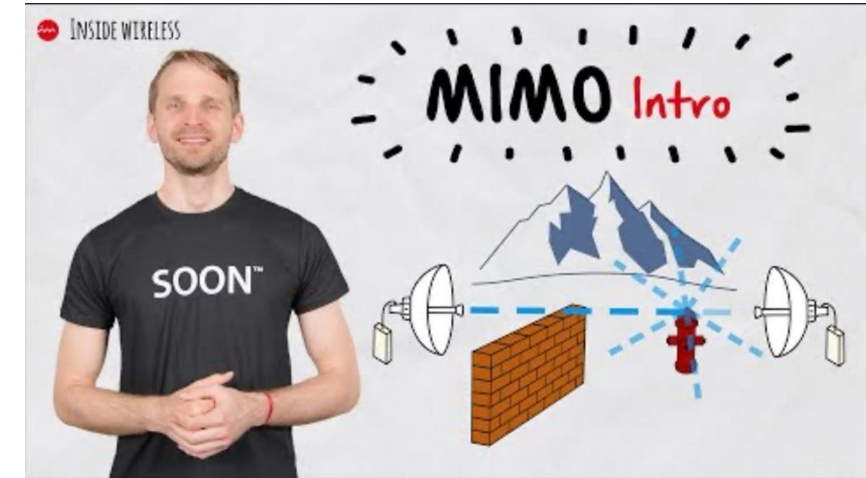
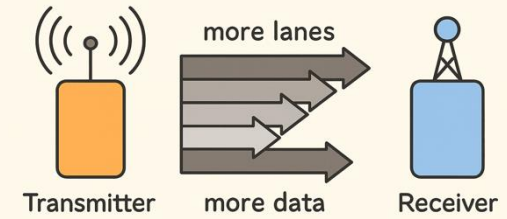
- With many antennas arranged in an array, the transmitter can change the
- phase of each antenna so that signals add up in a chosen direction and cancel
- in others. This creates a steerable “beam” of energy.

On Beamforming the transmitter changes their phase or amplitude to increase the receiver power.

Core Idea:

- MIMO adds more “lanes” of data.
- Beamforming points the energy where it matters.
- Together, they enable higher speed, range, and reliability.

MIMO & BEAMFORMING



How Radios Fix Bit Errors

Wireless signals are noisy. Fading, interference, and PA distortion all cause bit errors. Error-correction coding allows the receiver to detect and fix errors, making modern wireless reliable even with weak or noisy signals.

Why We Need It:

- RF channels corrupt bits (noise, multipath, interference)
- Raw BER is too high for QAM/OFDM
- Coding adds redundancy so the receiver can reconstruct the original data

Convolutional Codes + Viterbi Decoder

- One of the earliest error-correction systems used in GSM, Wi-Fi, satellites.
- Bits are encoded using shift registers that “mix” the input bits.
- The Viterbi algorithm finds the most likely original bit sequence.

Turbo Codes

- Used in 3G/4G LTE (2 convolutional encoders + interleaver)
- Very close to Shannon capacity

LDPC (Low-Density Parity Check Codes)

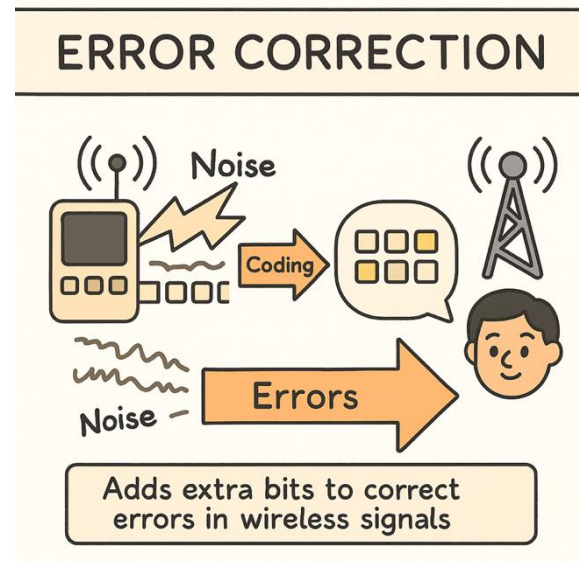
- Main error-correction code in 5G NR and Wi-Fi 6/6E
- Uses sparse parity-check matrices

How Coding Improves Reliability:

- Enables high-order QAM (64QAM, 256QAM, 1024QAM)
- Adds redundant bits (parity, checks) (Receiver uses math to detect and correct errors)
- Reduces required SNR for same BER

Key Idea:

- Error-correction adds smart redundancy so the radio can fix mistakes.
- Without coding, modern high-speed wireless systems would not work.
- **Coding adds smart redundancy so the receiver can repair errors.**



Software Defined Radio (SDRs)

Software-Defined Radio (SDR) is a radio architecture where most of the signal-processing traditionally done in hardware (mixers, filters, demodulators, modulators, equalizers) is moved into software running on a processor or FPGA.

Instead of building a new analog circuit for each protocol, an SDR converts the RF signal into digital samples as early as possible (near the antenna) and lets software decide:

- which modulation to use
- which band or channel to tune to
- how to filter the signal
- how to demodulate and decode it

Modern SDR systems use:

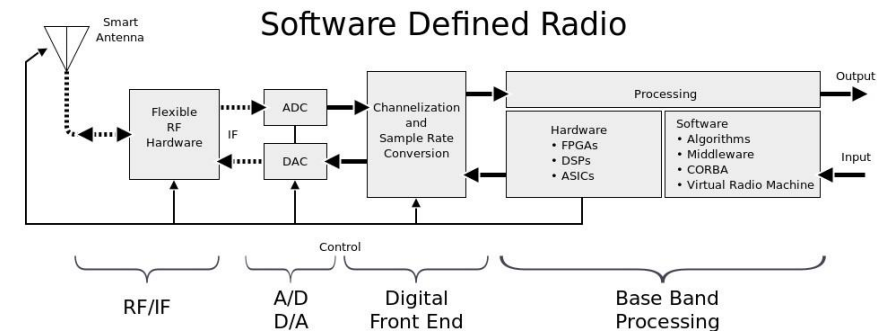
- A wideband RF front-end
- High-speed ADC/DAC converters
- Digital downconversion (DDC) and upconversion (DUC)
- FFT/IFFT for OFDM and channelization
- DSP algorithms for filtering, demodulation, and decoding

Why SDR Matters:

- One hardware platform can support many wireless standards (FM, AM, GSM, LTE, Wi-Fi, Bluetooth, 5G...)
- Easy upgrades: change the protocol by updating software
- Great for testing, prototyping, and research
- Enables cognitive radio, dynamic spectrum access, and modern 5G/6G systems

Key Idea:

- Analog hardware does only the minimum necessary (LNA, PA, mixers, filters).
- Everything else is flexible and programmable in software.



Test Equipment

Building and debugging RF systems requires specialized test equipment. These instruments allow you to **measure**, **characterize**, and **verify** that your RF circuits perform as designed. Here are the four fundamental tools every RF engineer uses.

Vector Network Analyzer (VNA)

A VNA measures how RF signals travel through a device by analyzing both **magnitude** and **phase** of signals. It's the most powerful tool for characterizing RF components. It measures the S-parameters, and Impedance matching views on Smith Chart

Spectrum Analyzer

A spectrum analyzer shows the **frequency content** of a signal. It displays **power vs. frequency** — essentially an FFT of the RF signal in real-time.

Signal Generator (RF Signal Generator / Synthesizer)

A signal generator creates precise RF signals at specific frequencies, power levels, and modulation types. It's the **signal source** for testing receivers, mixers, and systems. What it generates:

- **CW (Continuous Wave)**: Pure sine wave at one frequency
- **Modulated signals**: AM, FM, PM, QAM, OFDM, etc.
- **Arbitrary waveforms**: Custom I/Q data

Power Meter

A power meter measures the **absolute RF power** in Watts (or dBm) with high accuracy. It's the **reference standard** for power measurements. Common uses:

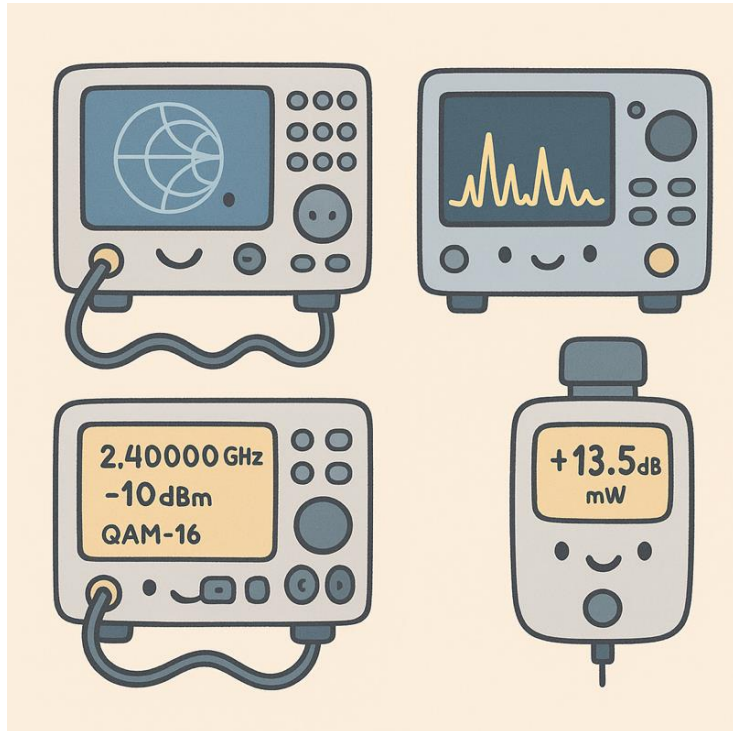
- Antenna input power
- Cable loss measurement
- Amplifier gain



Test Equipment (cont.)

These four instruments form the core of an RF lab:

- **VNA** → Characterization
- **Spectrum Analyzer** → Frequency analysis
- **Signal Generator** → Test stimulus
- **Power Meter** → Accurate power measurement



Task	VNA	Spectrum Analyzer	Signal Generator	Power Meter
Antenna match (S11)	✓	✗	✗	✗
Filter characterization	✓	✓	✓	✗
Find unwanted signals	✗	✓	✗	✗
Test receiver sensitivity	✗	✗	✓	⚠
Measure transmitter power	✗	⚠	✗	✓
Design matching networks	✓	✗	✗	✗

5G Part 1

5G is the latest generation of cellular technology, designed to deliver **ultra-high speeds, ultra-low latency**, and **massive device connectivity**. It represents a fundamental shift from previous generations, enabling not just faster phones, but entirely new use cases like autonomous vehicles, remote surgery, smart cities, and industrial IoT.

Three Main Use Cases

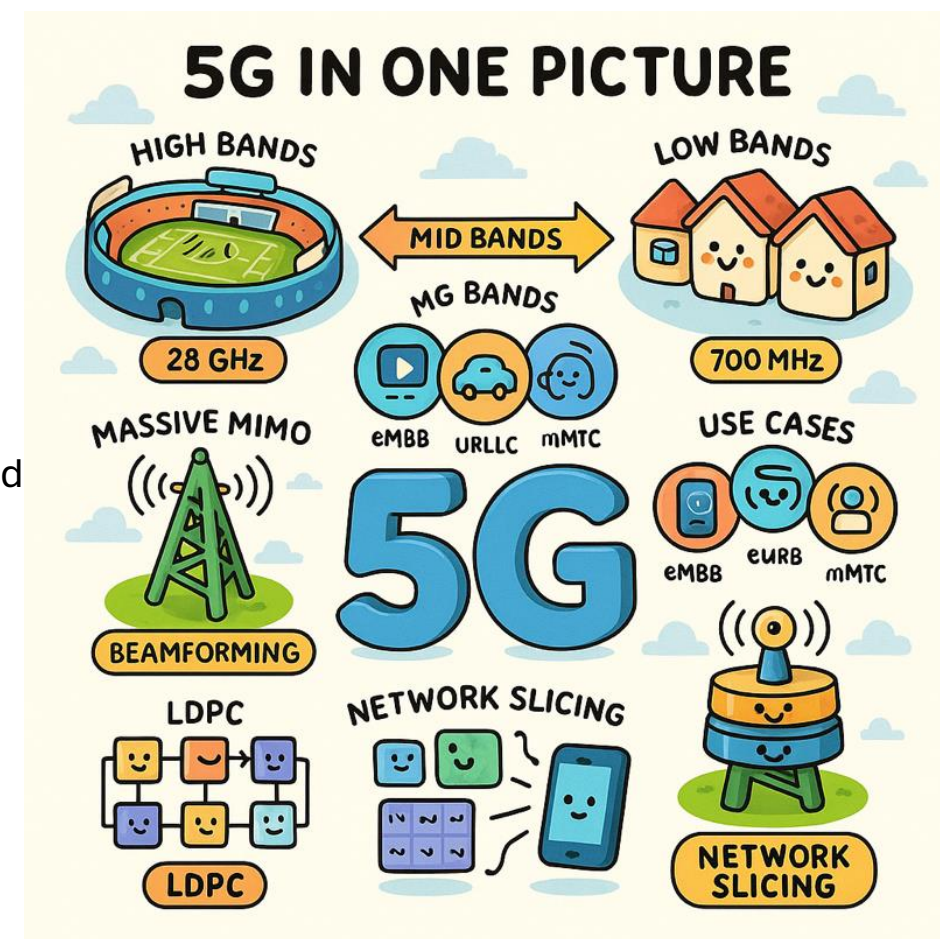
5G is designed around three distinct service categories:

1. eMBB (Enhanced Mobile Broadband)
2. URLLC (Ultra-Reliable Low-Latency Communications)
3. mMTC (Massive Machine-Type Communications)

Frequency Bands — Low, Mid, and High

5G operates across three distinct frequency ranges, each with different characteristics:

- Low-Band (Sub-1 GHz): "Coverage Layer"
- Mid-Band (1-6 GHz): "Capacity Layer" — The Sweet Spot
- High-Band / mmWave (24-100 GHz): "Ultra-High Speed Layer"



5G Part 2 (Key RF / PHY Technologies)

Massive MIMO + Beamforming

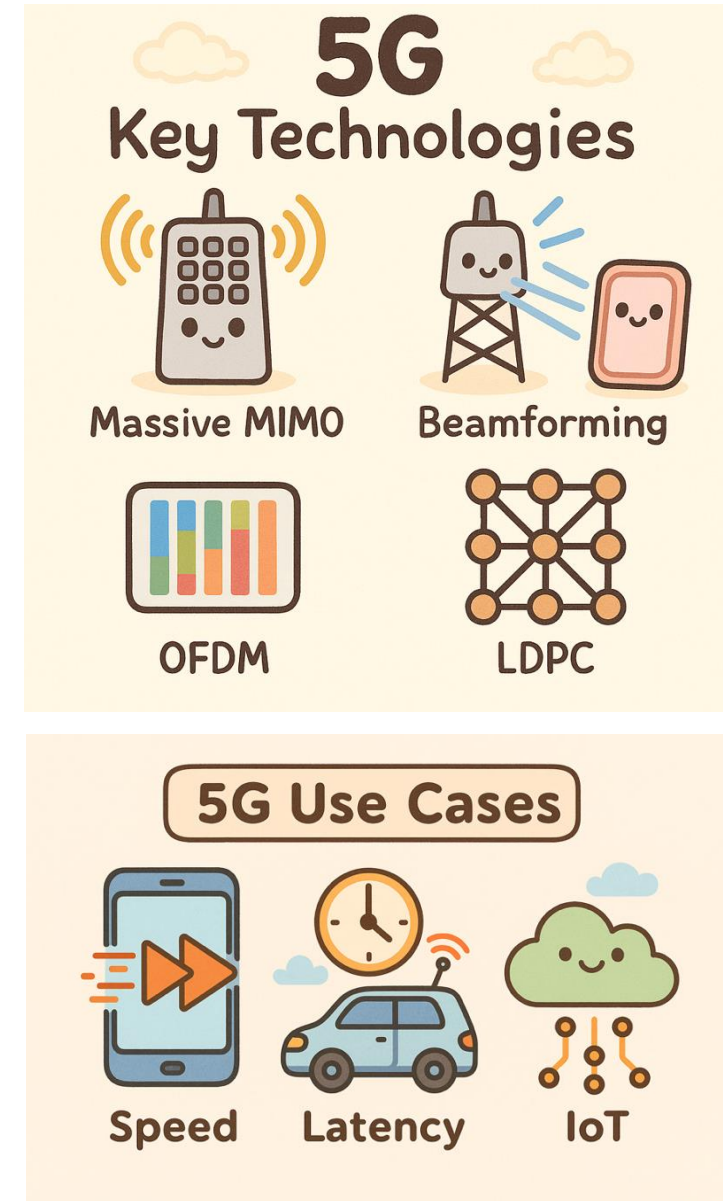
- Dozens of antennas at base station
- Steer narrow beams, reuse spectrum, fight path-loss

OFDM with flexible numerology

- Variable subcarrier spacing (15–240 kHz)
- Same waveform scales from low band to mmWave

High-order QAM + Advanced Coding

- Up to **256-QAM** in practice
- **LDPC (data)** and **Polar codes (control)** → near-Shannon performance



5G Part 3 (Frequency Bands — Coverage vs Capacity)

5G is deployed across three very different frequency layers. Each layer trades off **range**, **speed**, and **penetration**.

LOW-BAND (Sub-1 GHz) — Coverage Layer

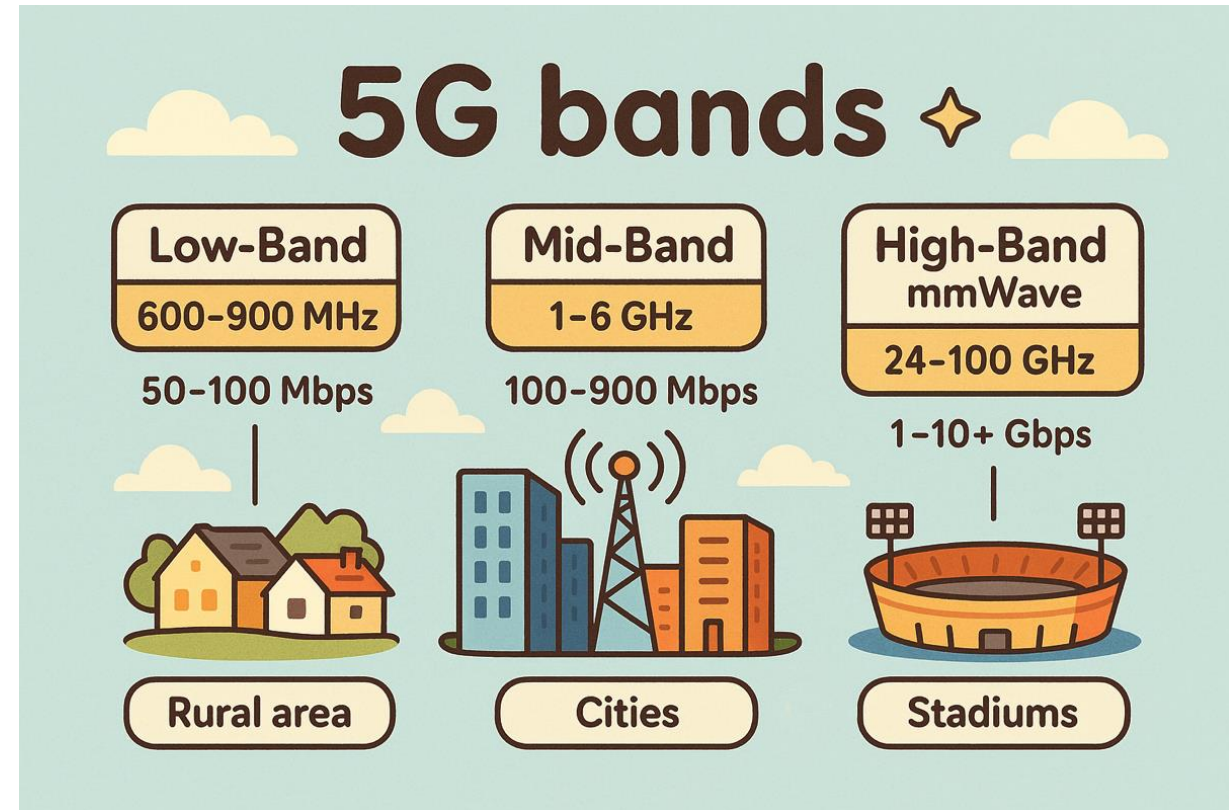
- 600–900 MHz
- Longest range (kilometers)
- Best wall penetration
- Lower speeds (50–150 Mbps)
- Ideal for rural, wide-area coverage

MID-BAND (1–6 GHz, esp. 3.5 GHz) — Capacity Layer

- 2.5 GHz, 3.5 GHz (C-band), 4.8 GHz
- Best balance of coverage & speed
- Hundreds of Mbps → 1+ Gbps
- Most important band for global 5G
- Urban + suburban deployment

HIGH-BAND (mmWave, 24–40+ GHz) — Hotspot Layer

- 24, 28, 39 GHz
- Multi-Gbps speeds (1–5+ Gbps)
- Very short range (100–300 m)
- Blocked by walls, trees, even hands
- Used in stadiums, airports, dense cities



5G Part 4 (Numerology)

“How 5G chooses the OFDM subcarrier spacing & timing.”

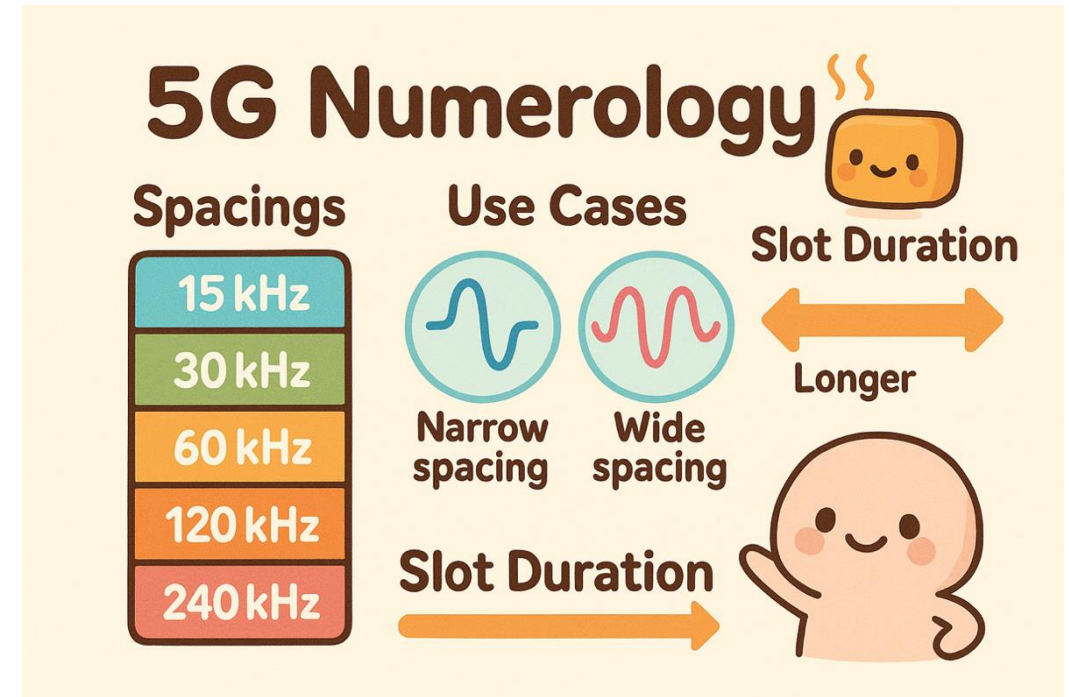
In 4G LTE, OFDM has **one fixed subcarrier spacing: 15 kHz** (always the same).

5G New Radio (NR) breaks this rule.

Instead, 5G introduces **multiple subcarrier spacings** that scale up or down depending on frequency band and latency needs.

This flexible system is called **numerology**.

5G numerology = flexible subcarrier spacing that adapts OFDM to different frequencies and use cases.



Numerology μ	Subcarrier Spacing Δf
0	15 kHz
1	30 kHz
2	60 kHz
3	120 kHz
4	240 kHz (mmWave)

References

- [Maxwell Equations](#), [Caltech Maxwell Video](#), [Focus on Maxwell Equations Forms](#), [MIT Math Heavy](#), [Hendrik Lorentz](#)
- [Understanding Electromagnetics Radiation](#), [Radio Waves](#)
- [All Modulation Types Explained in 3 Minutes](#)
- [How Radar Works](#)
- [What is QAM Modulation](#) (Inside QAM)
- [What is OFDM](#)
- [Visualizing Digital Modulation \(ASK, FSK, BPSK, DPSK, QPSK, QAM\)](#)
- [How to Get Phase From a Signal \(Using I/Q Sampling\)](#)
- [What is a Mixer? Modern RF and Microwave Mixers](#)
- [Channel Models in Wireless Communication](#)
- [What is Beamforming](#), [Another Tutorial on Beamforming](#)
- [Radio Antenna Theory 101](#), [Antenna Gain](#), [Basics of Antenna and Beamforming](#)
- [5G in 7 Minutes](#), [5G Network Architecture](#), [5G Overview Playlist](#), [5G Beamforming](#), [Cellular Communication Terminology](#)
- [5G Course Playlist](#), [5G Control and User Channels](#), [Error Corrections in 5G](#)
- [Open RAN \(ORAN\) Playlist](#), [Open5Gs](#), [Web](#) and [Docs](#)